

CompTIA Linux+ (XK0-005)

Access Control Utilities

- Introduction
 - Lab Topology
 - Exercise 1 - Controlling Access to Directory and Files with Standard Permissions
 - Exercise 2 - Controlling Access to Directory and Files with Advanced Permissions
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Introduction

XK0-005

File Permissions

Setfacl

Getfacl

Sticky Bit

Welcome to the **Access Control Utilities** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Controlling Access to Directory and Files with Standard Permissions
- Exercise 2 - Controlling Access to Directory and Files with Advanced Permissions

After completing this module, you should be able to:

- Create Files and View Permissions
- Modify File Permissions
- Configure Permissions using Setfacl and Create Inheritable Permission
- Set Sticky Bit to Control Access

Exam Objectives

The following exam objectives are covered in this module:

2.5 Given a scenario, apply the appropriate access controls

- Command-line utilities

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABSRV01** - Ubuntu 20.04 - Stand-Alone Ubuntu Server (Console)
- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)
- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)
- **PLABWIN10** - Windows 10 - Stand-Alone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Controlling Access to Directory and Files with Standard Permissions

In this exercise, you will view and modify standard permissions and control ownership. You will be working with the `chmod`, `chown` and `chgrp` commands.

Learning Outcomes

After completing this exercise, you should be able to:

- Create Files and View Permissions
- Modify File Permissions

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Create Files and View Permissions

In this task, you will create files and view permissions.

Step 1

Connect to **PLABCLIENT01**.

Login as **administrator** using the following **Password**:

Passw0rd

Press **Enter**.

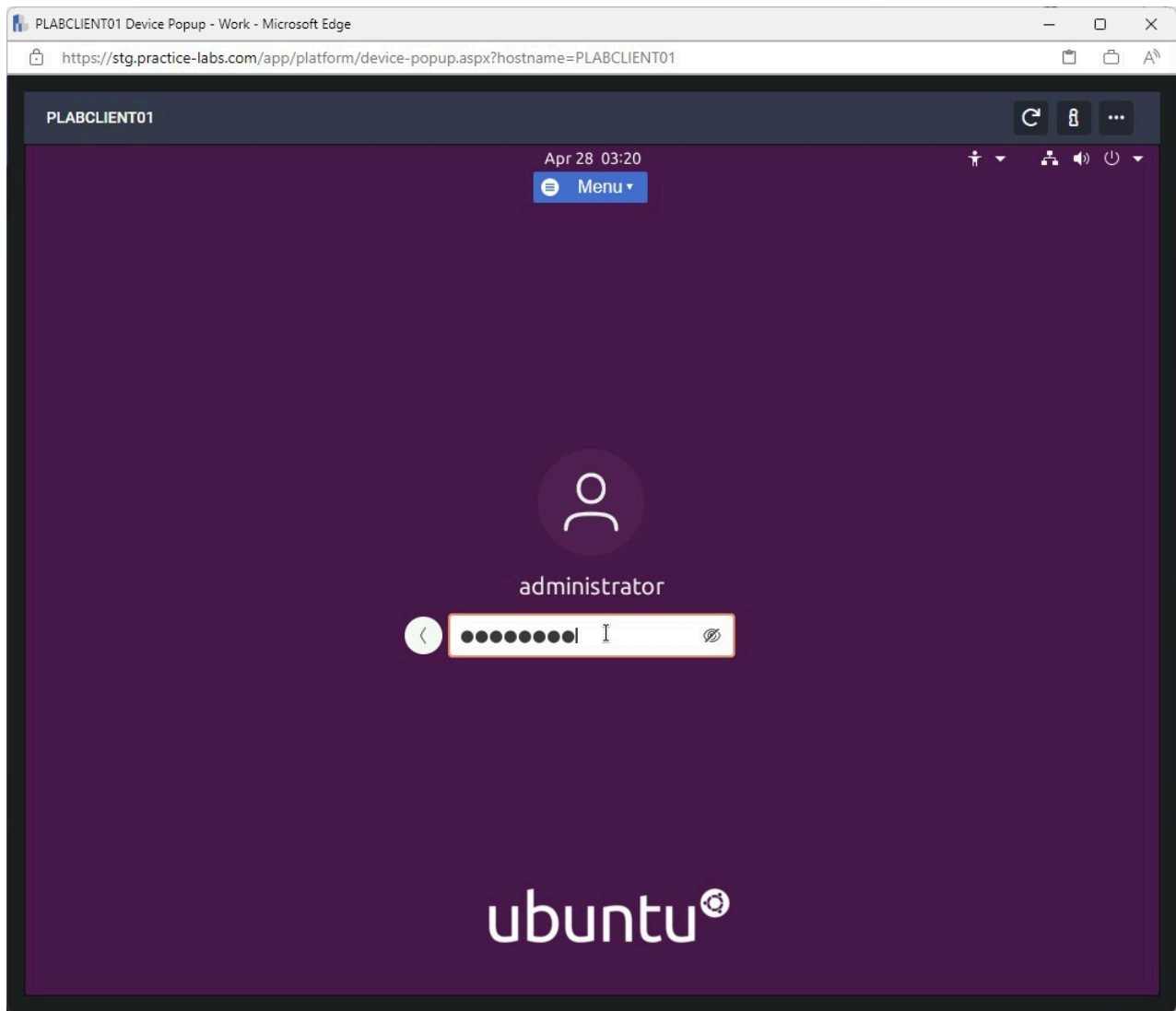


Figure 1.1 Screenshot of PLABCLIENT01: The login screen is displayed, showing the required credentials typed in.

Step 2

Click the **Show applications** icon on the bottom left corner.

Scroll down and select **Terminal**.

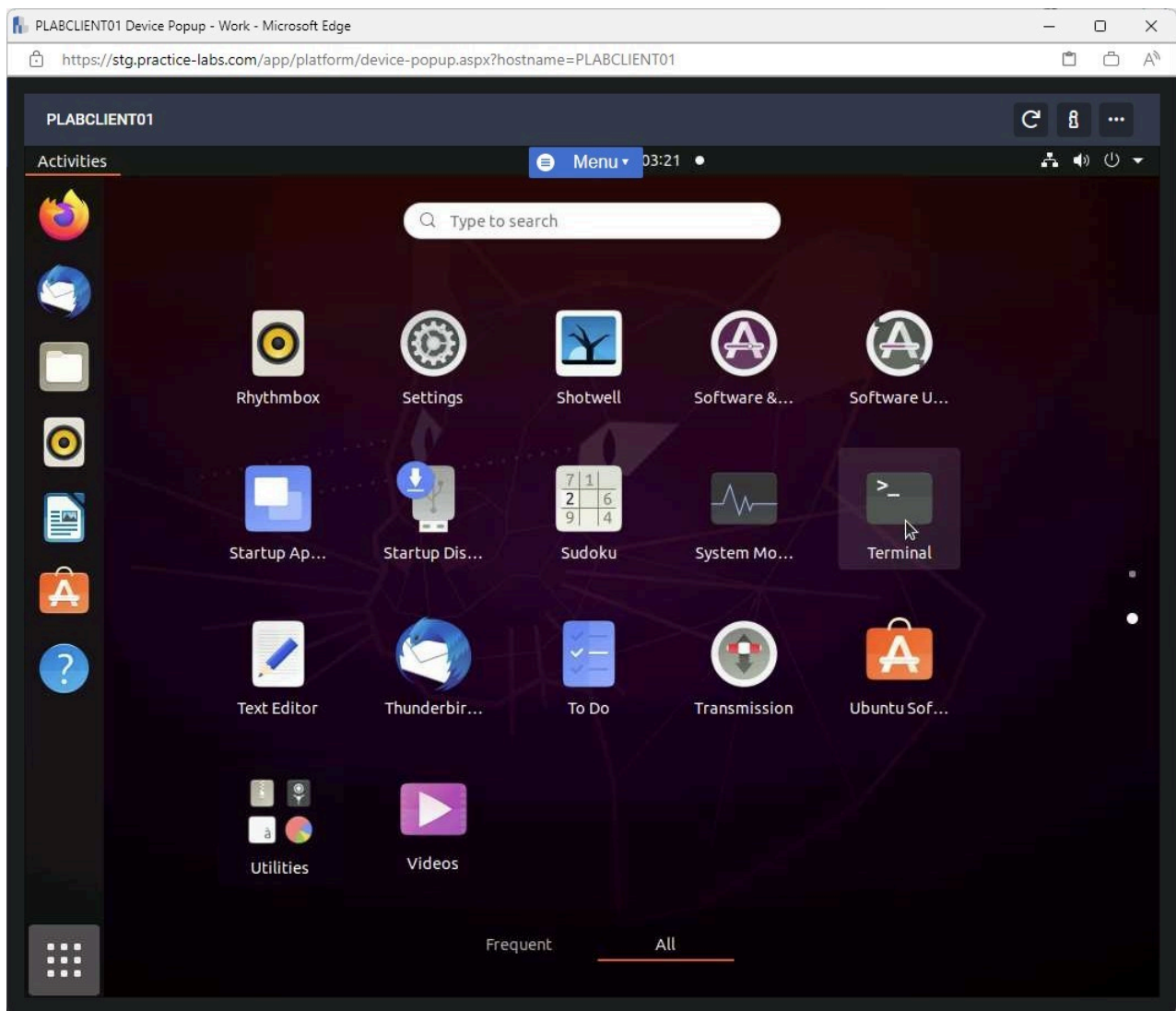


Figure 1.2 Screenshot of PLABCLIENT01: Show applications icon is clicked, and Terminal is selected.

Step 3

In the **Terminal** window, type the following command to navigate to the **Documents** folder:

```
cd Documents/
```

Press **Enter**.

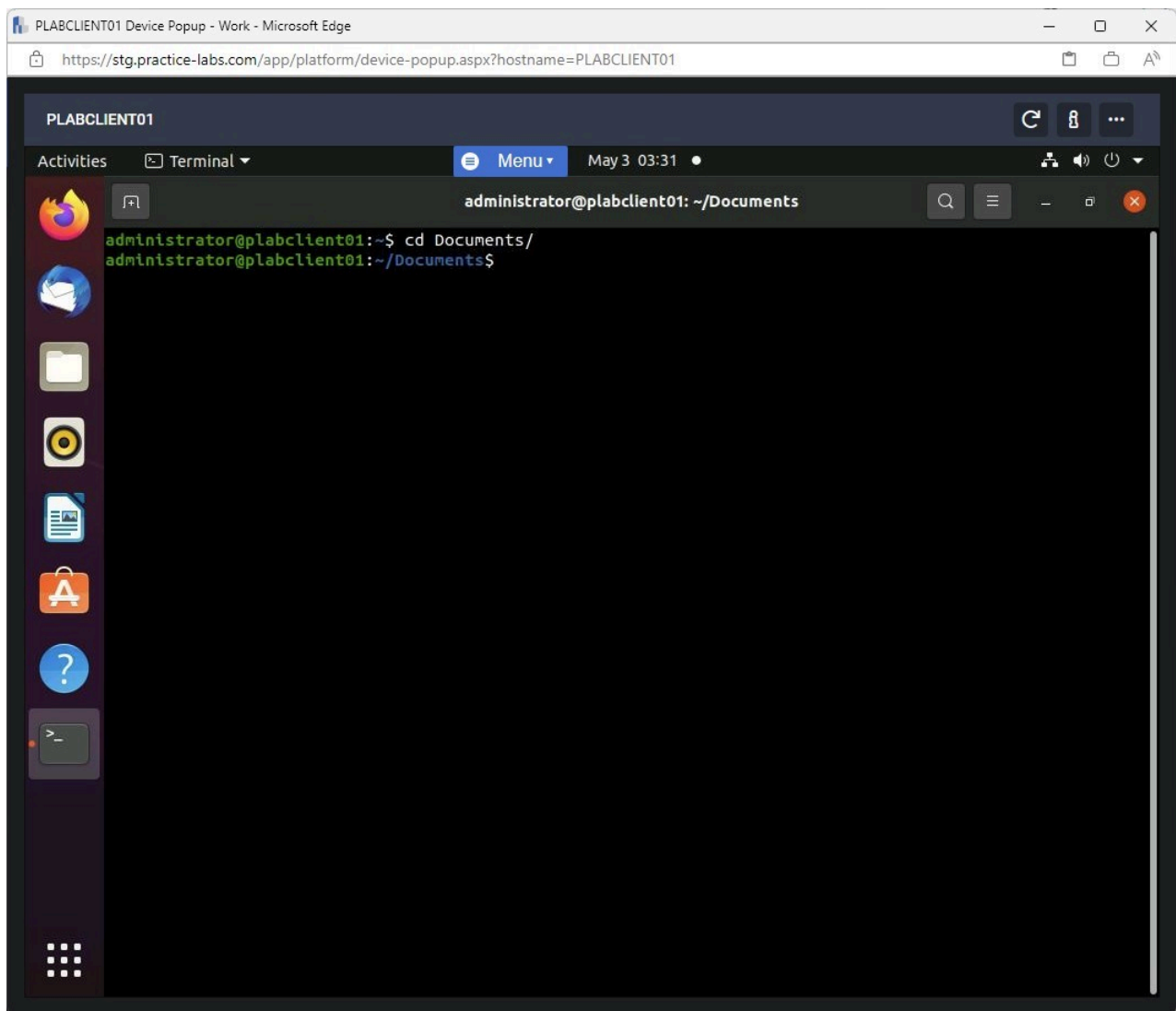


Figure 1.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 4

In the **Terminal** window, type the following commands to create 3 files (Press **Enter** after each command):

```
touch textfile1.txt  
touch textfile2.txt  
touch textfile3.txt
```

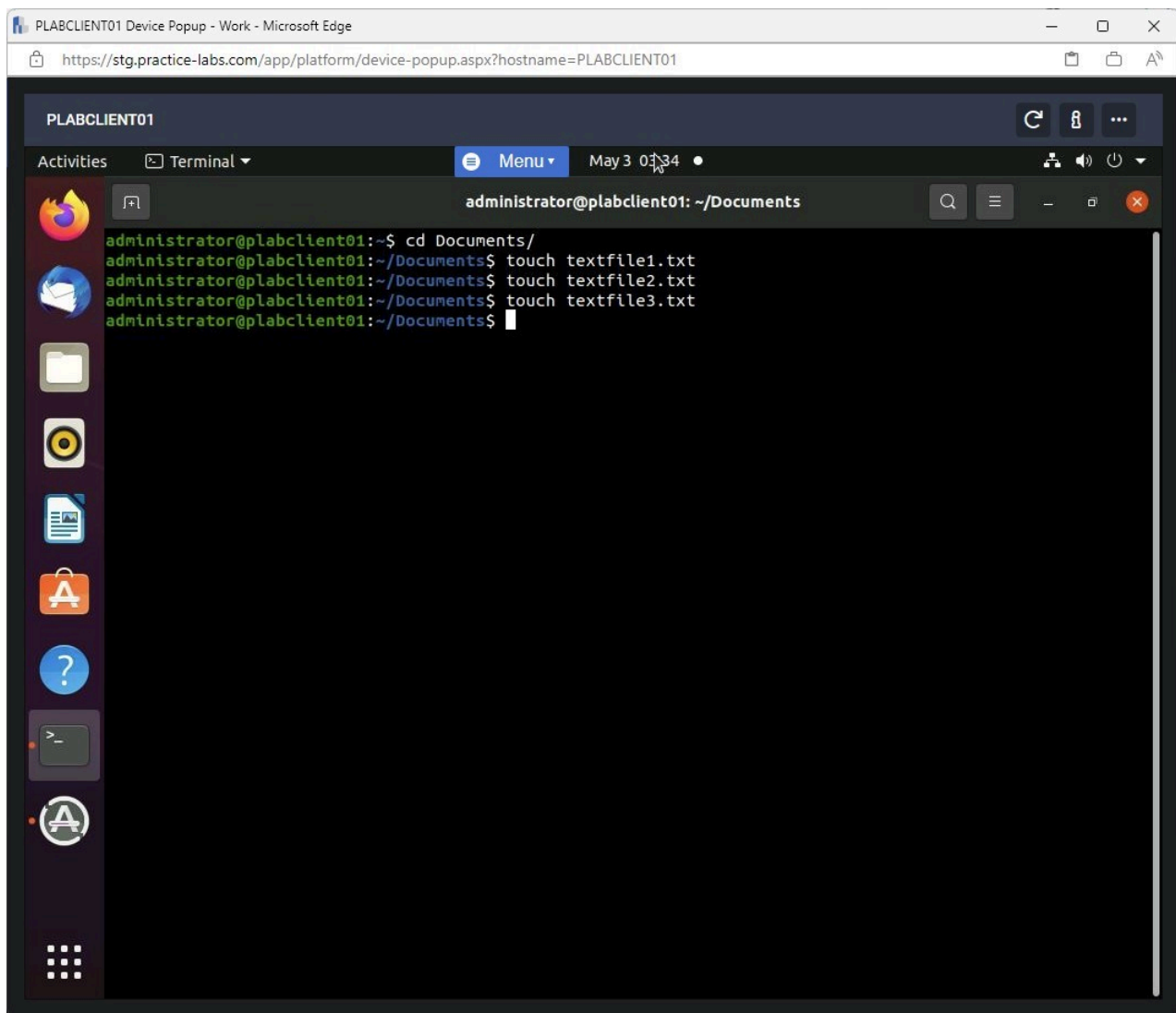


Figure 1.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
ls
```

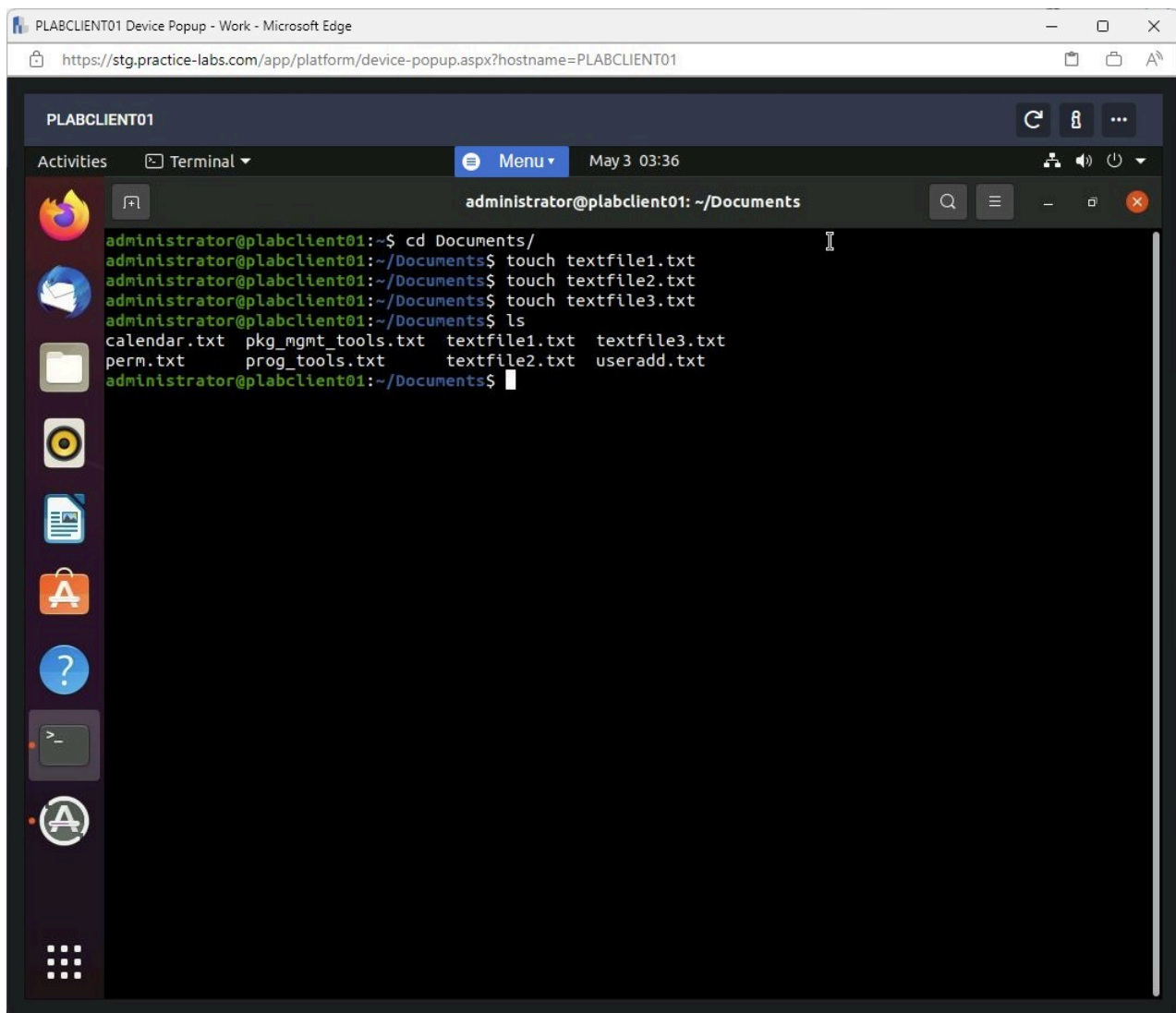


Figure 1.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the 3 files are now created in the **Document** folder.

Step 6

In the **Terminal** window, type the following command to navigate to the **etc** directory:

```
cd /etc
```

Press **Enter**.

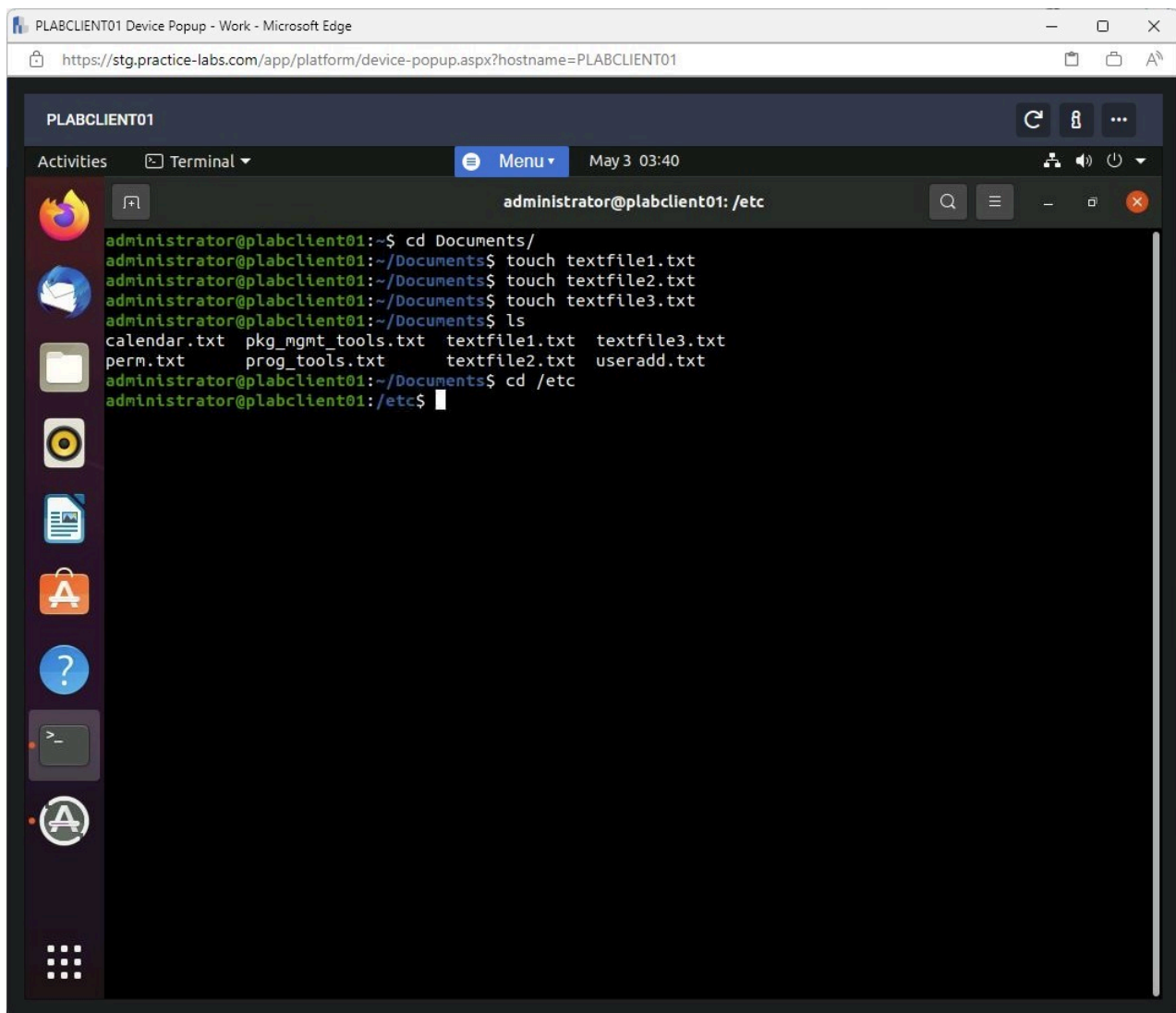


Figure 1.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *etc* directory contains all the Linux system configuration files.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
cat passwd
```

The screenshot shows a terminal window titled 'PLABCLIENT01' with the URL 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal displays the following commands and output:

```
administrator@plabclient01:~/Documents$ touch textfile3.txt
administrator@plabclient01:~/Documents$ ls
calendar.txt  pkg_mgmt_tools.txt  textfile1.txt  textfile3.txt
perm.txt      prog_tools.txt      textfile2.txt  useradd.txt
administrator@plabclient01:~/Documents$ cd /etc
administrator@plabclient01:/etc$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:/:/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115:/:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
```

Figure 1.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *cat* (short for concatenate) command will display the contents of the */etc/passwd* file. Notice you are also able to view the permissions.

Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
cat shadow
```

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 3 03:57
administrator@plabclient01: /etc

www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:nonexistent:/usr/sbin/nologin
syslog:x:104:110:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115:nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
saned:x:117:123:/var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,,:/run/hplip:/bin/false
whoopsie:x:120:125:nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:123:128:PulseAudio daemon,,,:/var/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:124:65534:/run/gnome-initial-setup:/bin/false
gdm:x:125:130:Gnome Display Manager:/var/lib/gdm3:/bin/false
sssd:x:126:131:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
administrator:x:1000:1000:administrator,,,:/home/administrator:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
administrator@plabclient01:/etc$ cat shadow
cat: shadow: Permission denied
administrator@plabclient01:/etc$

```

Figure 1.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: You get the **Permission denied** error. This is because only the root users and users that belong to the shadow group have access to the `/etc/shadow` files.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l shadow
```

The screenshot shows a terminal window titled 'PLABCLIENT01' with the prompt 'administrator@plabclient01: /etc'. The terminal displays a list of system users and their configurations. The command 'cat shadow' is entered, resulting in a 'Permission denied' message. Subsequently, the command 'ls -l shadow' is entered, displaying the file permissions and ownership of the shadow file.

```

list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115:/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin
saned:x:117:123:/var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,:/run/hplip:/bin/false
whoopsie:x:120:125:/nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:123:128:PulseAudio daemon,,:/var/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:124:65534:/run/gnome-initial-setup:/bin/false
gdm:x:125:130:Gnome Display Manager:/var/lib/gdm3:/bin/false
sssd:x:126:131:SSSD system user,,:/var/lib/sss:/usr/sbin/nologin
administrator:x:1000:1000:administrator,,:/home/administrator:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
administrator@plabclient01:/etc$ cat shadow
cat: shadow: Permission denied
administrator@plabclient01:/etc$ ls -l shadow
-rw-r----- 1 root shadow 1469 Mar 21  2022 shadow
administrator@plabclient01:/etc$

```

Figure 1.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The permissions for the shadow file is displayed:

- 1st character - “d” or “-” directory or file
- 1st column - “rw-r-----” = permissions (9 characters)
- 2nd column - “1” = # of hardlinks associated
- 3rd column - “root” = user (owner)
- 4th column - “shadow” = group
- 5th column - “1469” = size
- 6th column - “Mar 21 2022” = last modified date
- File name

Step 10

In the **Terminal** window, type the following command and press **Enter**:

clear

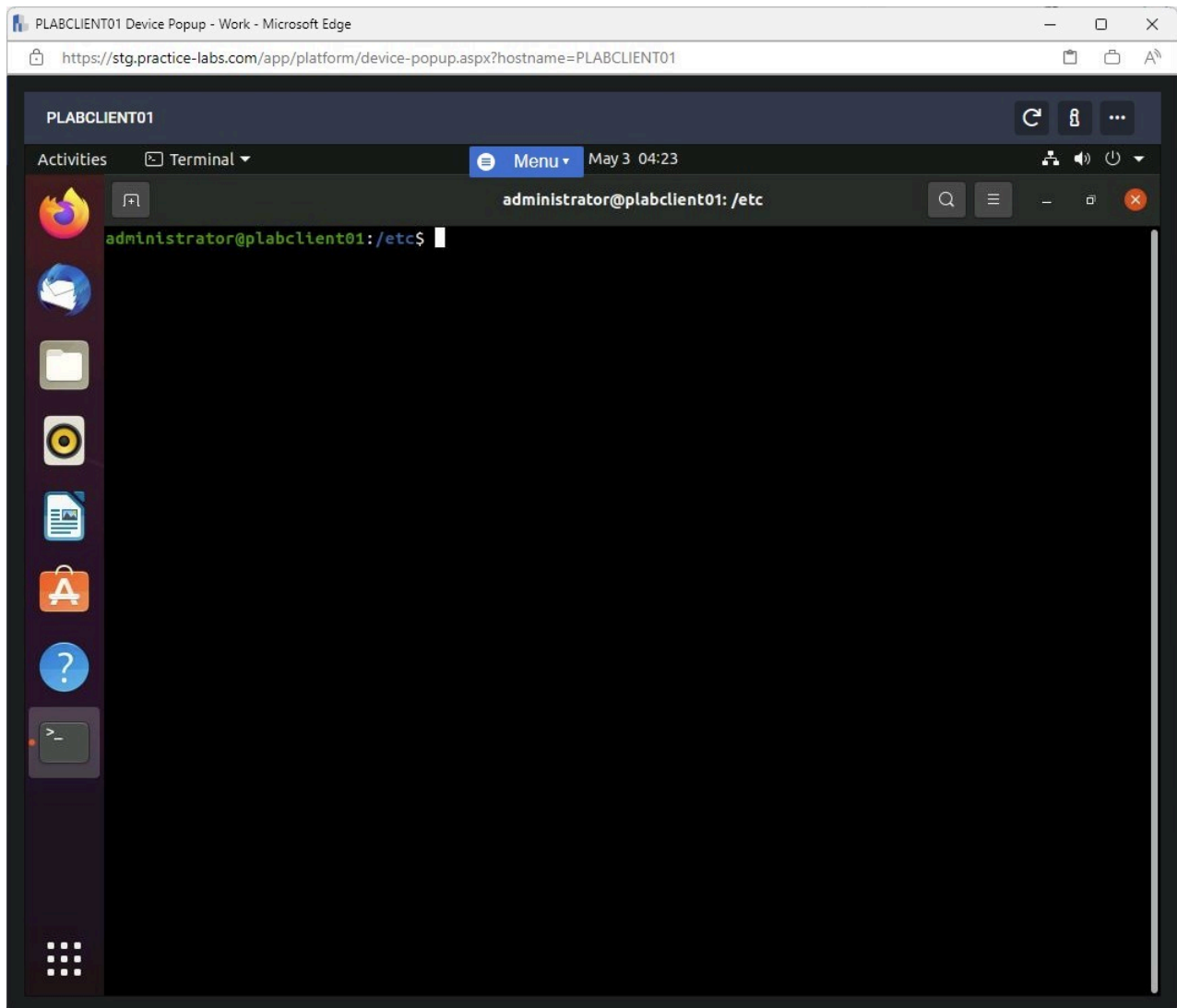


Figure 1.10 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *clear* command will clear the Terminal window.

Keep the **Terminal** window open.

Task 2 - Modify File Permissions

In this task, you will modify permissions as well as control ownership. You will use the `chmod`, `chown`, and `chgrp` commands on files and directories.

Step 1

Ensure you are connected to **PLABCLIENT01**, where the **Terminal** window is open.

In the **Terminal** window, type the following command to navigate to the **Documents** folder:

```
cd /home/administrator/Documents
```

Press **Enter**.

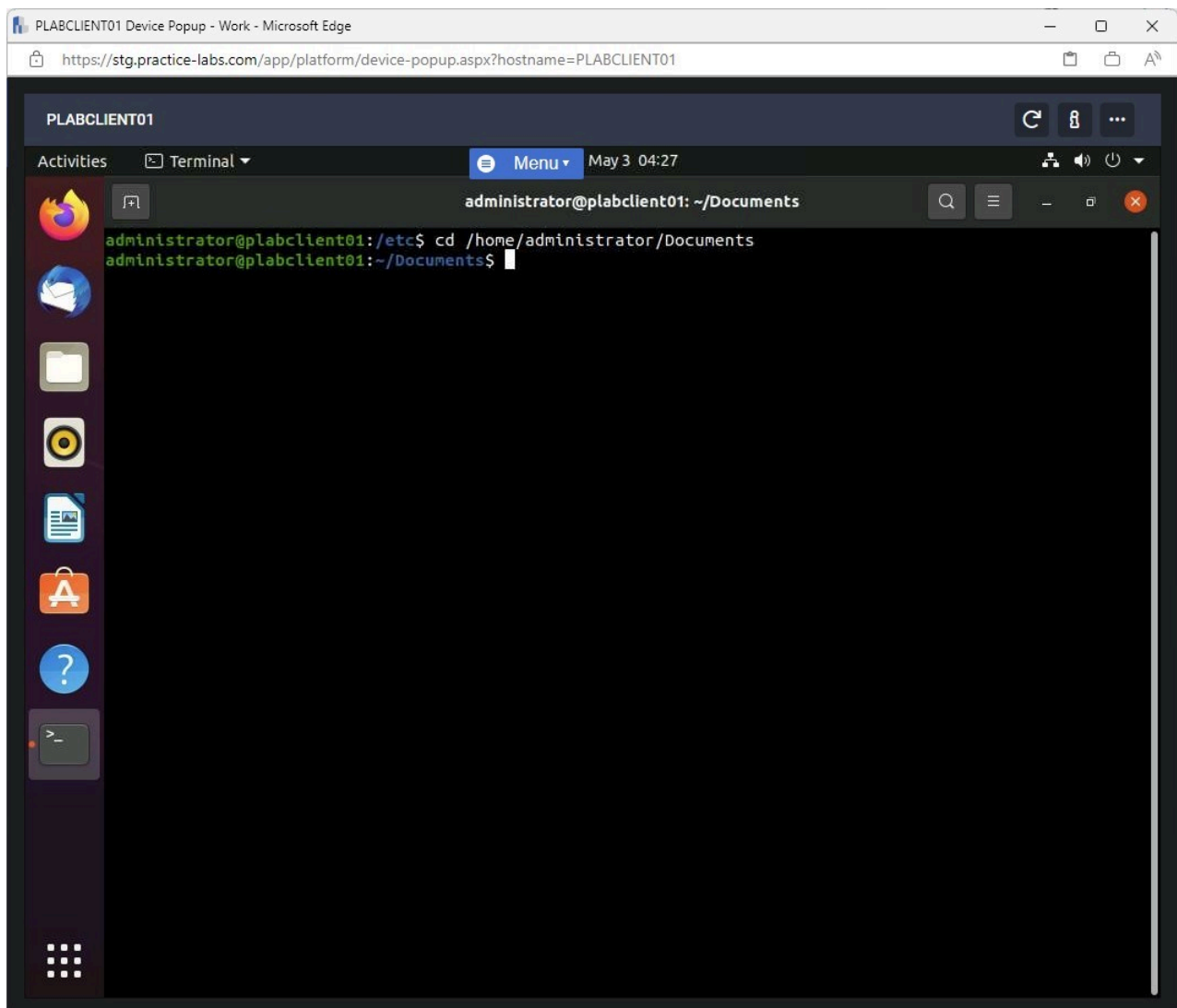


Figure 1.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
chmod 400 textfile1.txt
```

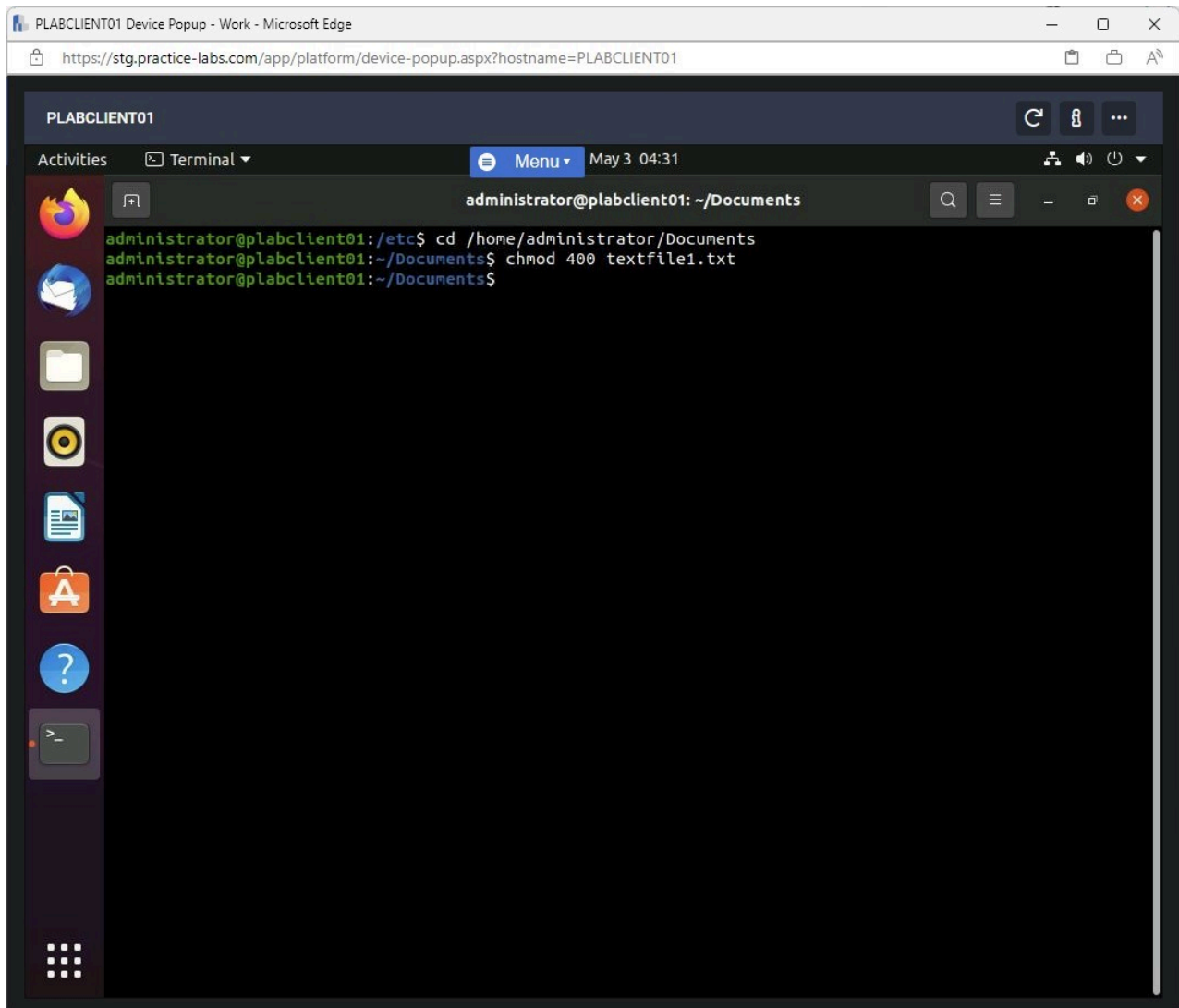


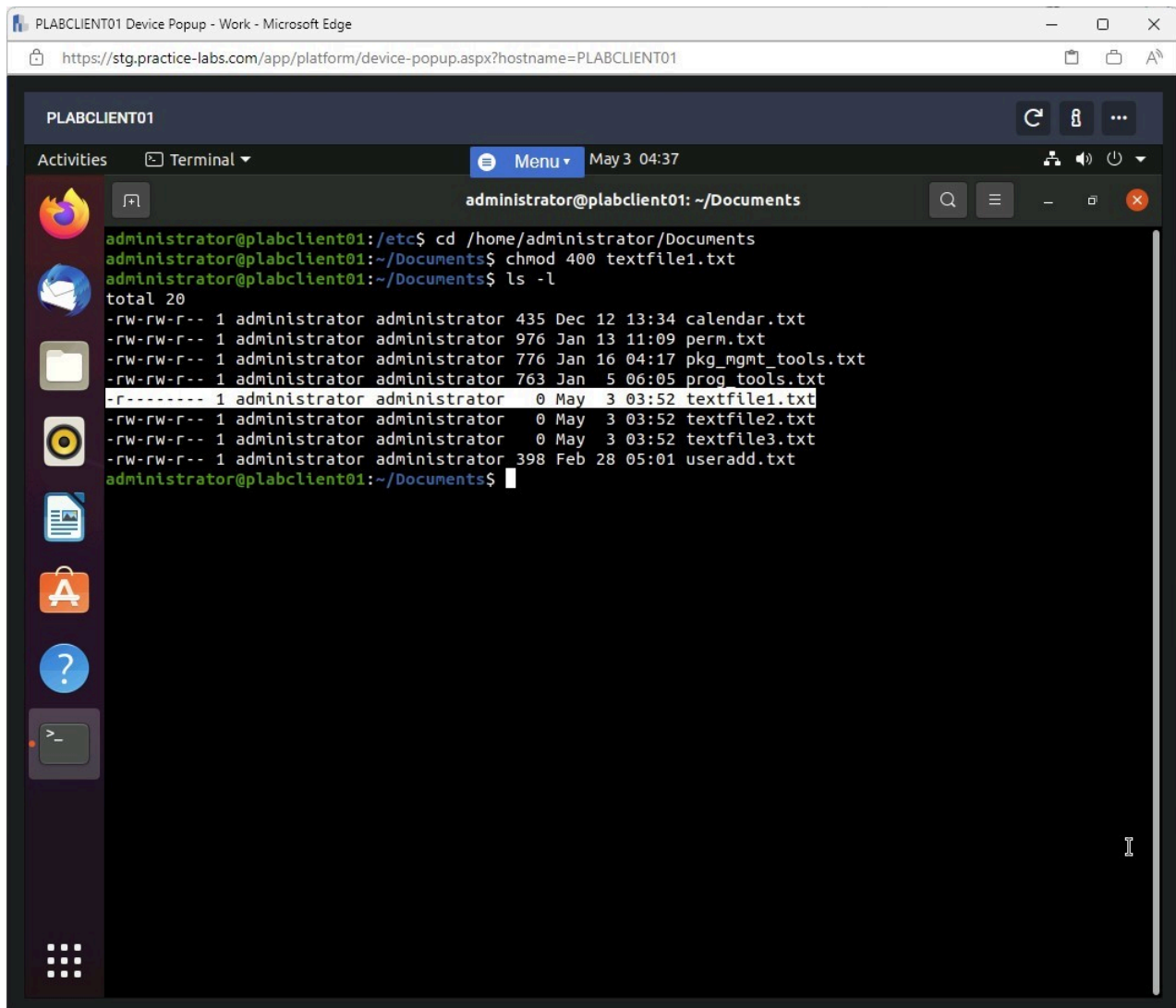
Figure 1.12 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **chmod** (change mode) command is used to modify the permissions of a file. The **chmod** command, when used with the **400** parameter, will give the file read only permission. 400 is its octal representation.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a menu bar and a sidebar. The terminal output shows the user 'administrator' at 'plabclient01' in the directory '~/Documents'. The user has executed the command 'ls -l', which displays a long listing of files in the directory. The files listed are: 'calendar.txt', 'perm.txt', 'pkg_mgmt_tools.txt', 'prog_tools.txt', 'textfile1.txt', 'textfile2.txt', 'textfile3.txt', and 'useradd.txt'. The permissions for 'textfile1.txt' are shown as '-r-----', indicating that only the owner has read permission.

```
administrator@plabclient01: /etc$ cd /home/administrator/Documents
administrator@plabclient01:~/Documents$ chmod 400 textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-r----- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.13 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice *textfile1.txt* now only has read permission.

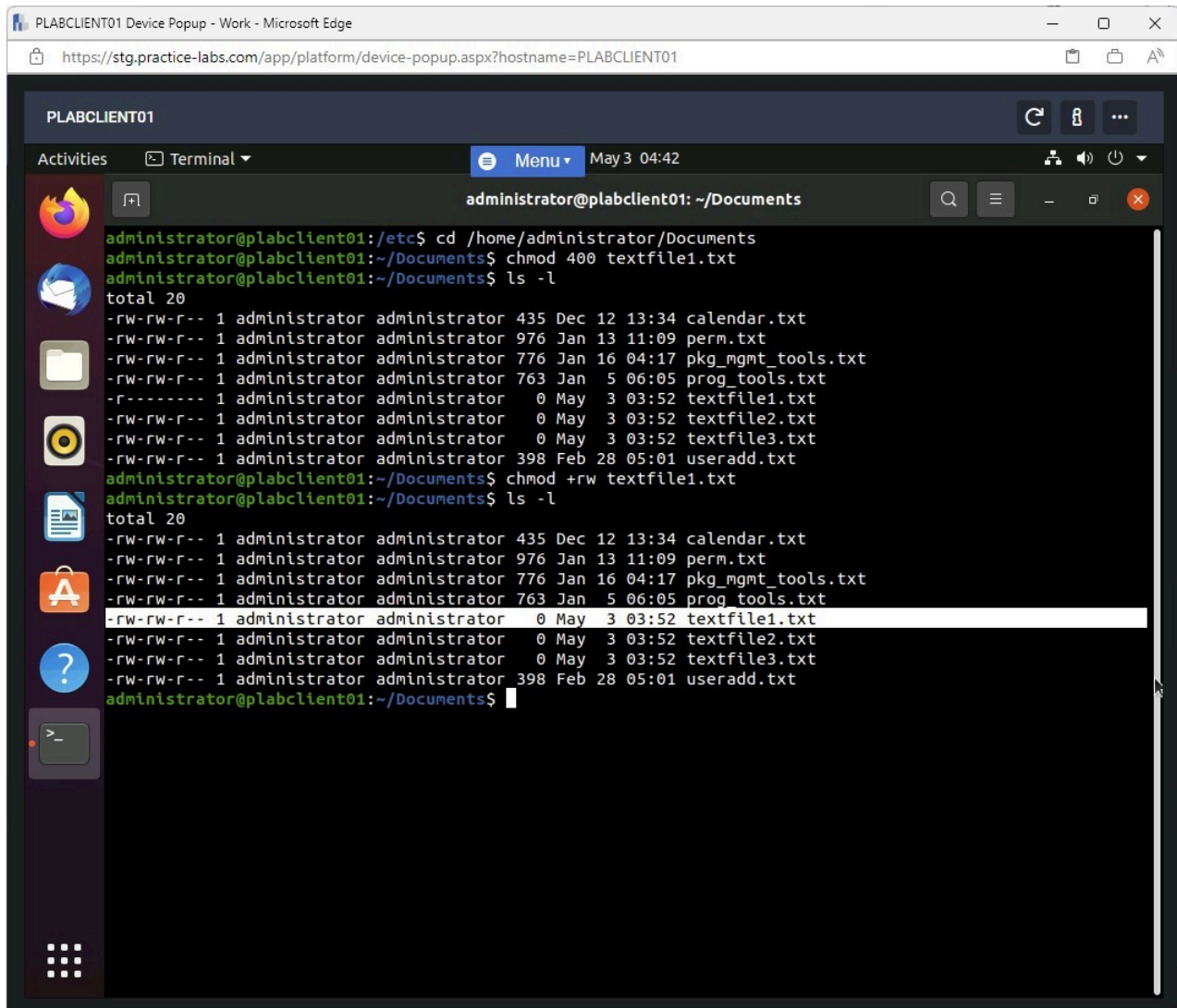
Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
chmod +rw textfile1.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The prompt is 'administrator@plabclient01: ~/Documents'. The user has entered the command 'ls -l', and the output is displayed. The output shows a list of files with their permissions, owner, group, size, and date. The file 'textfile1.txt' is highlighted in blue, showing its permissions as '-rw-rw-r--'. The user has also entered the command 'chmod 400 textfile1.txt' and 'ls -l' again, but the output is not fully visible.

```
administrator@plabclient01: /etc$ cd /home/administrator/Documents
administrator@plabclient01:~/Documents$ chmod 400 textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-r----- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ chmod +rw textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.14 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the permission of **textfile1.txt** has now changed. The permissions are as follows:

: The file type. In this case, '-' denotes a file.

rw : Read and write permissions for the owner of the file.

rw : Read and write permissions for members in the group having access to the file.

r : Read permission for all other users.

Execute permissions are denoted by an 'x'.

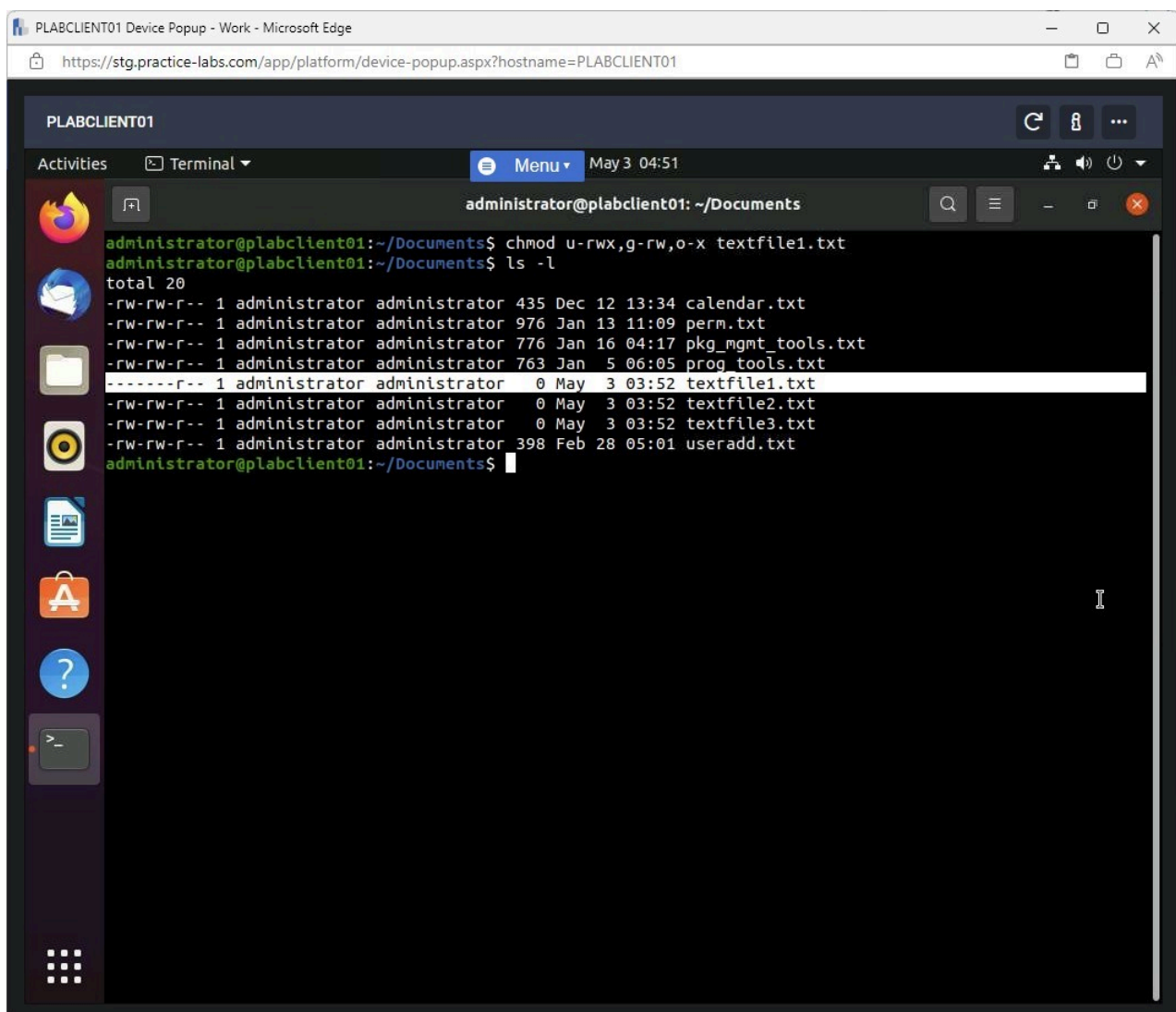
Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
chmod u-rwx,g-rw,o-x textfile1.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01 Device Popup - Work - Microsoft Edge". The terminal is running on a system named "PLABCLIENT01". The user is "administrator" and the current directory is "~/Documents". The terminal shows the following commands and output:

```
administrator@plabclient01: ~/Documents
administrator@plabclient01:~/Documents$ chmod u-rwx,g-rw,o-x textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.15 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **chmod** command used with the **u-rwx,g-rw,o-x** parameter will remove all permissions from the owner of the file and will give read permission to all other users.

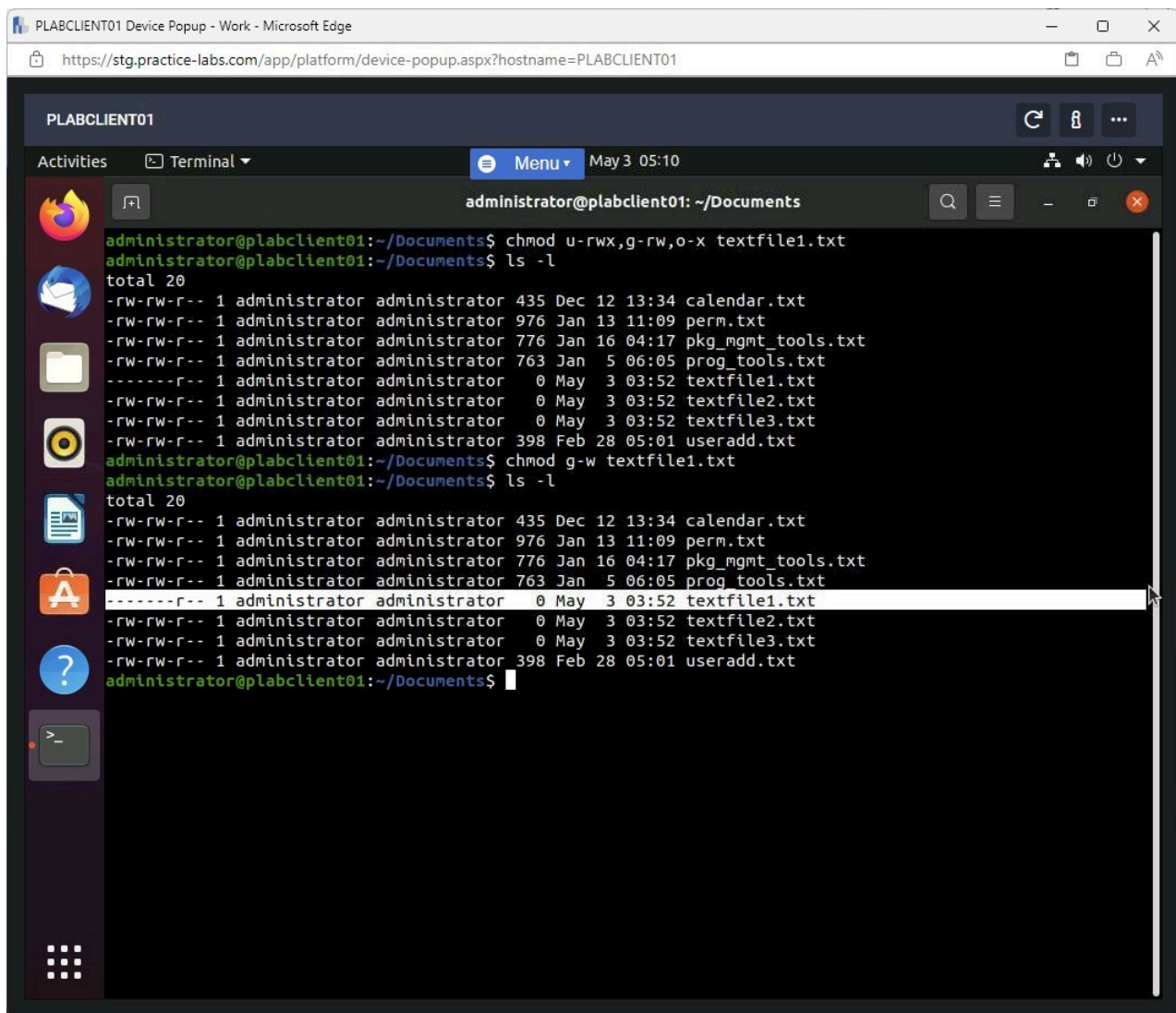
Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
chmod g-w textfile1.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 3 05:10
administrator@plabclient01: ~/Documents
administrator@plabclient01:~/Documents$ chmod u-rwx,g-rw,o-x textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ chmod g-w textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.16 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *chmod* command used with the *g-w* parameter will also remove all permissions from the owner of the file.

Step 7

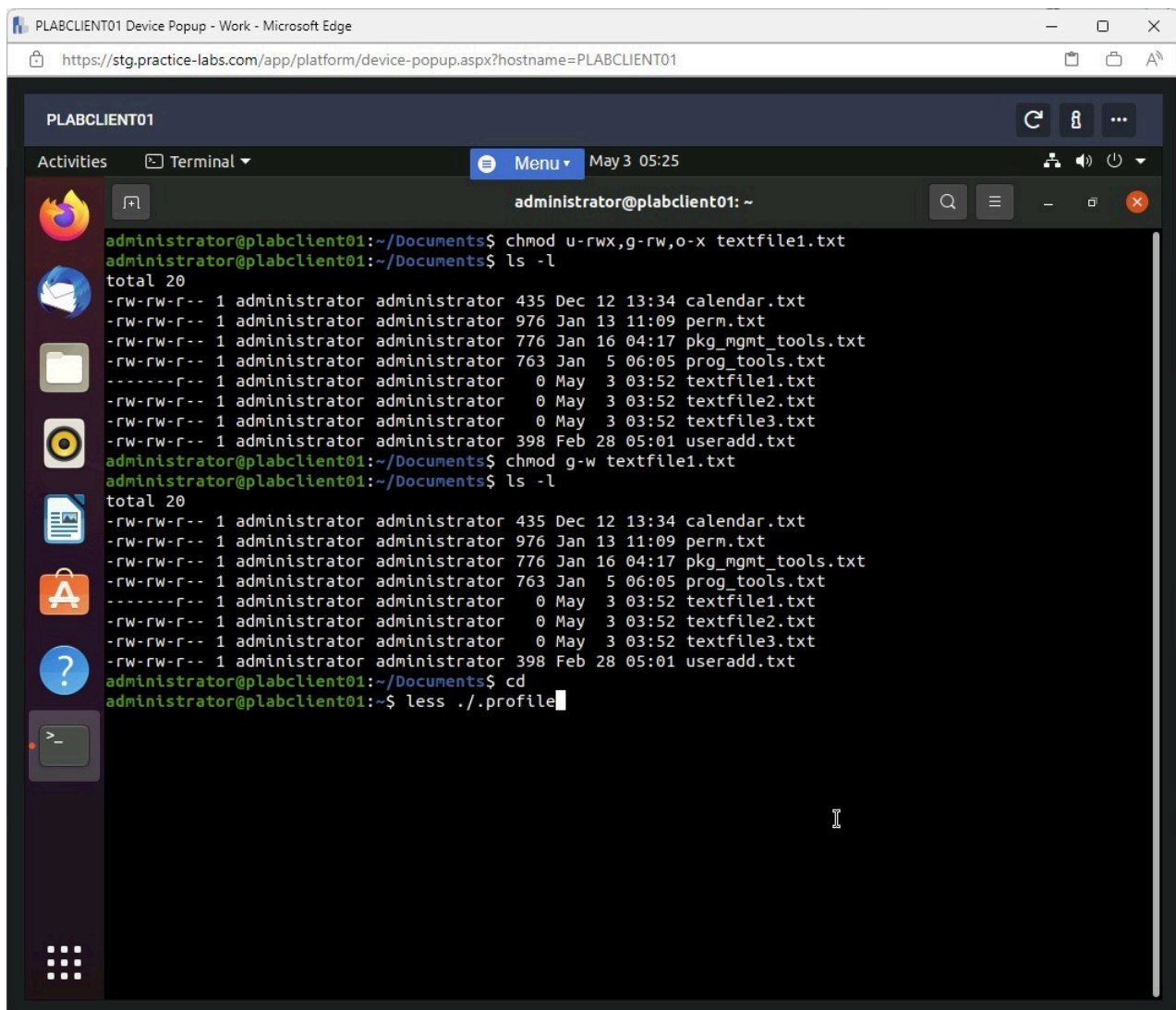
In the **Terminal** window, type the following command to navigate to the default directory:

```
cd
```

Press **Enter**.

Next, type the following command and press **Enter**:

```
less ../.profile
```



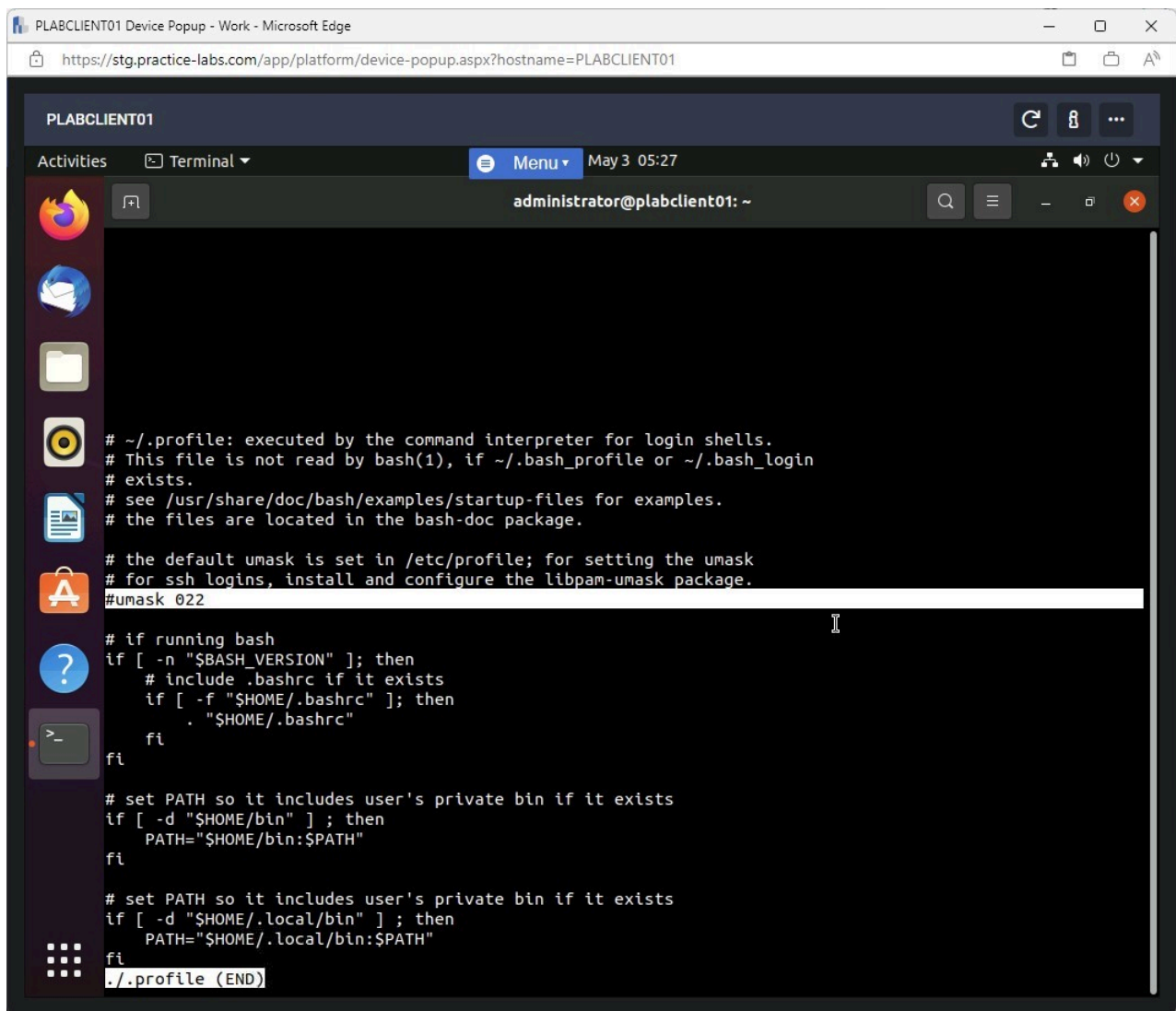
The screenshot shows a terminal window titled "PLABCLIENT01" with a menu bar and a sidebar. The terminal output is as follows:

```
administrator@plabclient01:~/Documents$ chmod u-rwx,g-rw,o-x textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ chmod g-w textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ cd
administrator@plabclient01:~$ less ../.profile
```

Figure 1.17 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 8

The output of the **less ../.profile** command is displayed.



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 3 05:27
administrator@plabclient01: ~

# ~/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
umask 022

# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
    if [ -f "$HOME/.bashrc" ]; then
        . "$HOME/.bashrc"
    fi
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/bin" ] ; then
    PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ] ; then
    PATH="$HOME/.local/bin:$PATH"
fi

./ .profile (END)
```

Figure 1.18 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the output of the command.

Note: The **less ~/.profile** command displays the default permissions as defined by the Linux system. Notice the line **umask 022**. The 022 octal is the inverse of the numeric code for permissions. For example, a umask of 022 is actually 755.

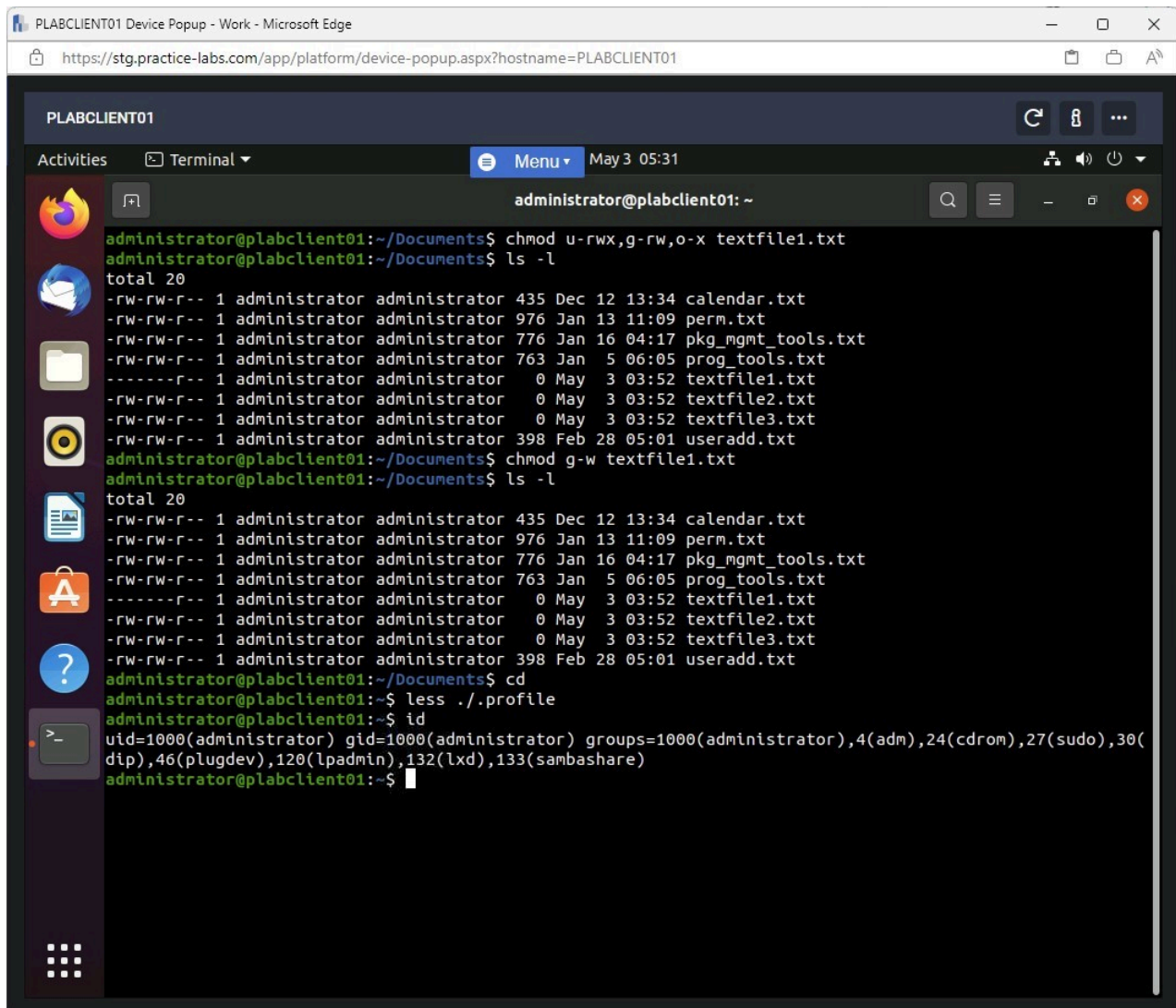
Step 9

Type the following to exit out of the **less** window:

q

In the **Terminal** window, type the following command and press **Enter**:

id



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 3 05:31
administrator@plabclient01: ~

administrator@plabclient01:~/Documents$ chmod u-rwx,g-rw,o-x textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ chmod g-w textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ cd
administrator@plabclient01:~$ less ../profile
administrator@plabclient01:~$ id
uid=1000(administrator) gid=1000(administrator) groups=1000(administrator),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),132(lxd),133(sambashare)
administrator@plabclient01:~$
```

Figure 1.19 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **id** command will list the profile permissions.

Step 10

In the **Terminal** window, type the following command and press **Enter**:

```
sudo useradd jdoe
```

Type the following **Password** and press **Enter**:

Passw0rd

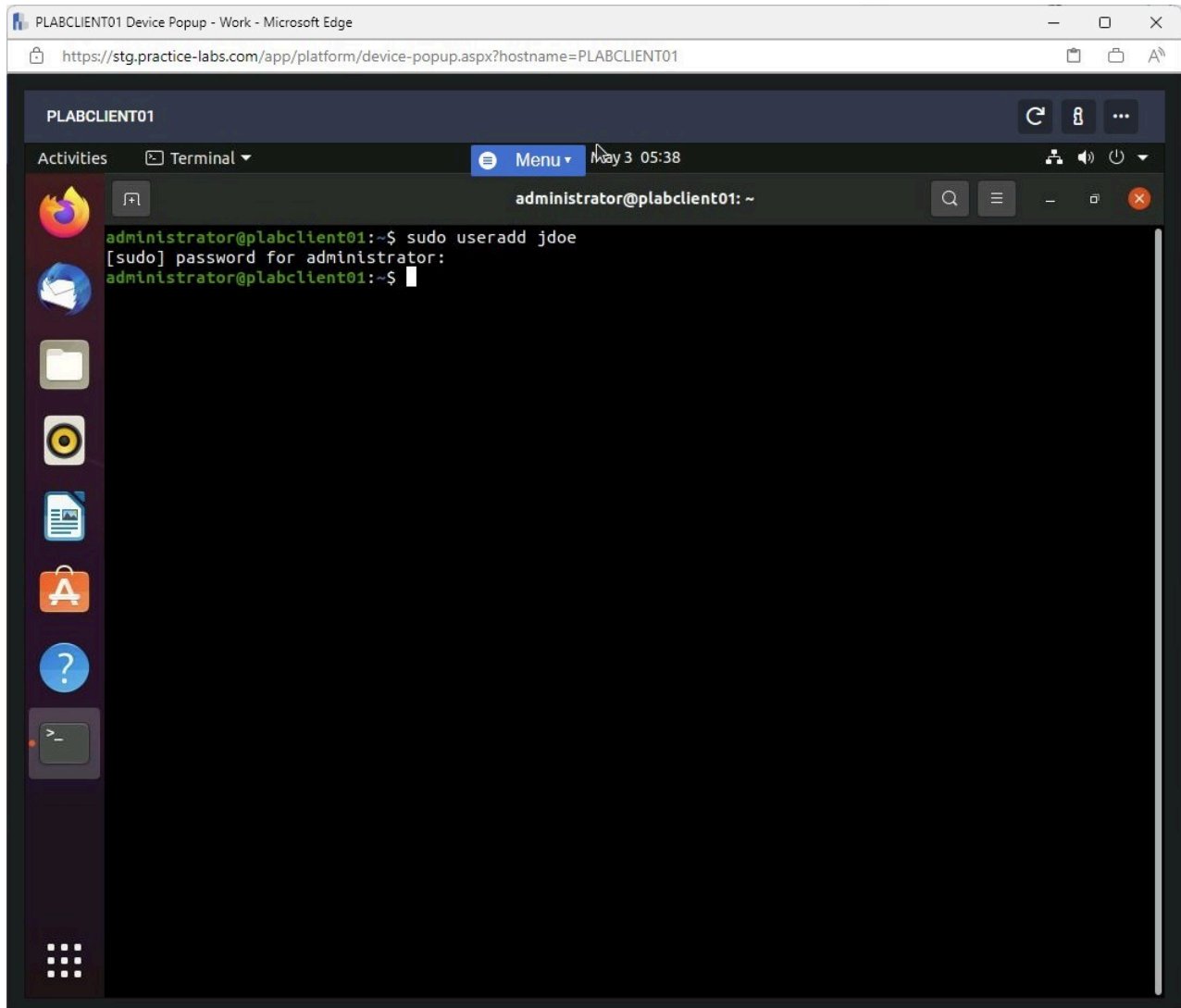


Figure 1.20 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above command will create a new user named **jdoe**.

Step 11

In the **Terminal** window, type the following command to navigate to the **Documents** directory:

```
cd Documents/
```

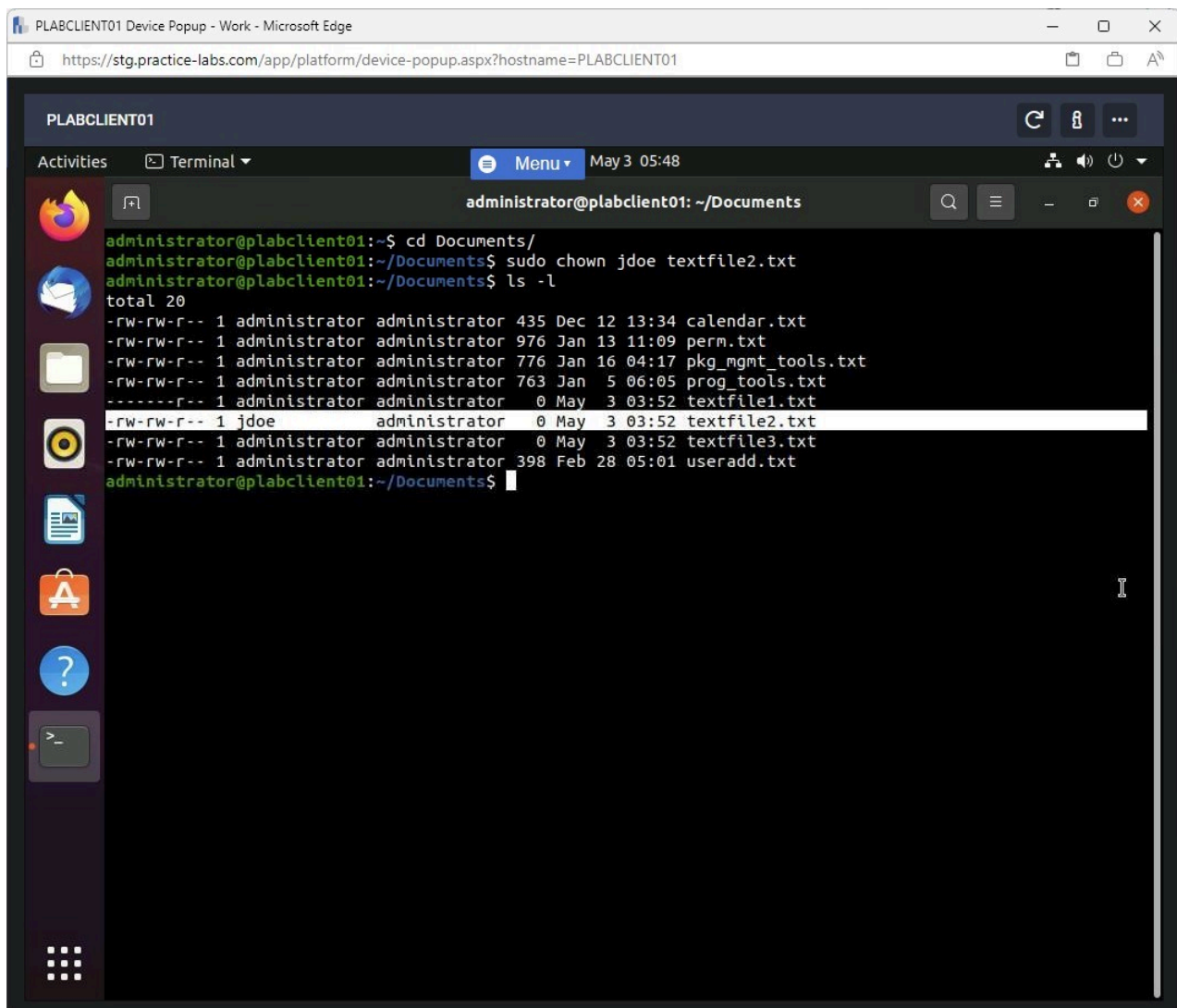
Press **Enter**.

Then, type the following command and press **Enter**:

```
sudo chown jdoe textfile2.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01". The terminal displays the following commands and output:

```
administrator@plabclient01: ~/Documents
administrator@plabclient01:~$ cd Documents/
administrator@plabclient01:~/Documents$ sudo chown jdoe textfile2.txt
administrator@plabclient01:~/Documents$ ls -l
```

Permissions	Link Count	User	Group	Size	Month	Day	Time	File Name
-rw-rw-r--	1	administrator	administrator	435	Dec	12	13:34	calendar.txt
-rw-rw-r--	1	administrator	administrator	976	Jan	13	11:09	perm.txt
-rw-rw-r--	1	administrator	administrator	776	Jan	16	04:17	pkg_mgmt_tools.txt
-rw-rw-r--	1	administrator	administrator	763	Jan	5	06:05	prog_tools.txt
-----r--	1	administrator	administrator	0	May	3	03:52	textfile1.txt
-rw-rw-r--	1	jdoe	administrator	0	May	3	03:52	textfile2.txt
-rw-rw-r--	1	administrator	administrator	0	May	3	03:52	textfile3.txt
-rw-rw-r--	1	administrator	administrator	398	Feb	28	05:01	useradd.txt

```
administrator@plabclient01:~/Documents$
```

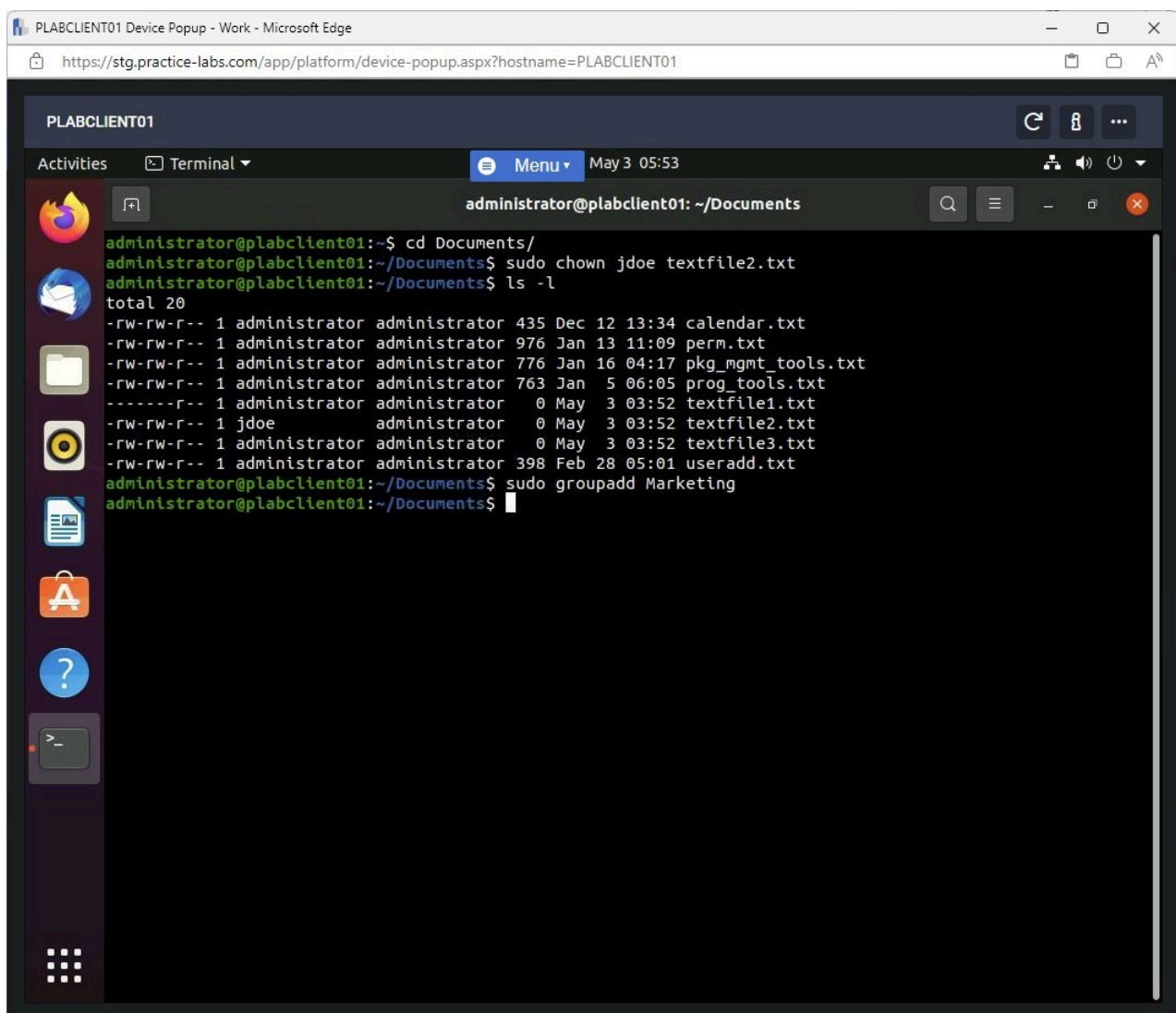
Figure 1.21 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **chown** command will change the ownership of the file. Notice **jdoe** is now the owner of **textfile2.txt**.

Step 12

In the **Terminal** window, type the following command and press **Enter**:

```
sudo groupadd Marketing
```



The screenshot shows a terminal window titled "PLABCLIENT01" with a dark background. The prompt is "administrator@plabclient01: ~/Documents". The user has entered the command "sudo groupadd Marketing" and pressed Enter. The terminal output shows the command being executed successfully. The terminal also displays the output of "ls -l" showing a list of files in the current directory, including "calendar.txt", "perm.txt", "pkg_mgmt_tools.txt", "prog_tools.txt", "textfile1.txt", "textfile2.txt", "textfile3.txt", and "useradd.txt". The file "textfile2.txt" is owned by "jdoe".

```
administrator@plabclient01:~$ cd Documents/
administrator@plabclient01:~/Documents$ sudo chown jdoe textfile2.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 jdoe administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo groupadd Marketing
administrator@plabclient01:~/Documents$
```

Figure 1.22 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above command will create a new group named **Marketing**.

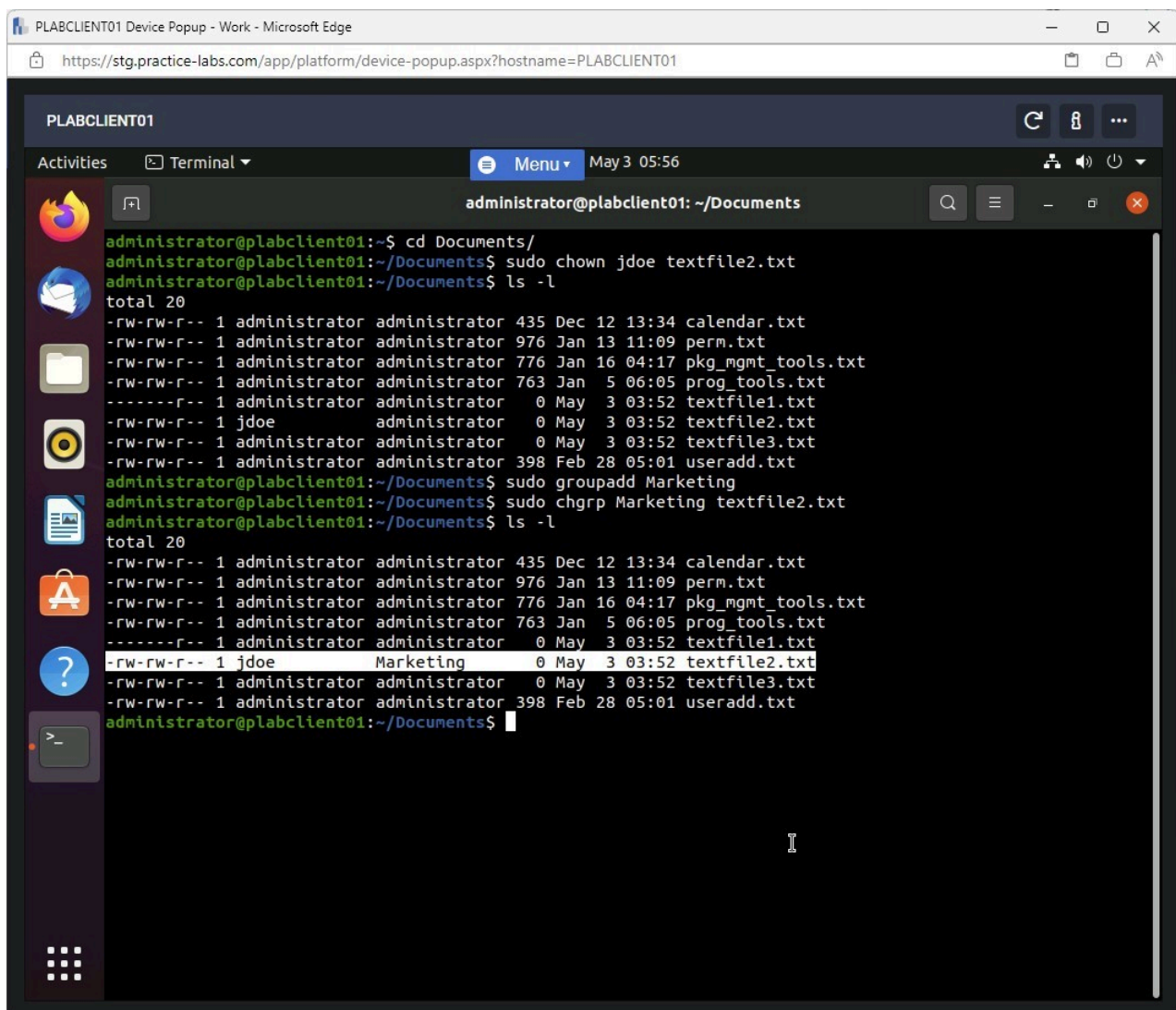
Step 13

In the **Terminal** window, type the following command and press **Enter**:

```
sudo chgrp Marketing textfile2.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01". The terminal is running as "administrator@plabclient01: ~/Documents". The commands and output are as follows:

```
administrator@plabclient01:~$ cd Documents/
administrator@plabclient01:~/Documents$ sudo chown jdoe textfile2.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 jdoe administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo groupadd Marketing
administrator@plabclient01:~/Documents$ sudo chgrp Marketing textfile2.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 jdoe Marketing 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.23 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *chgrp* command will change the group of *textfile2.txt* to *Marketing*.

Step 14

In the **Terminal** window, type the following command and press **Enter**:

```
sudo chown :Marketing textfile3.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```

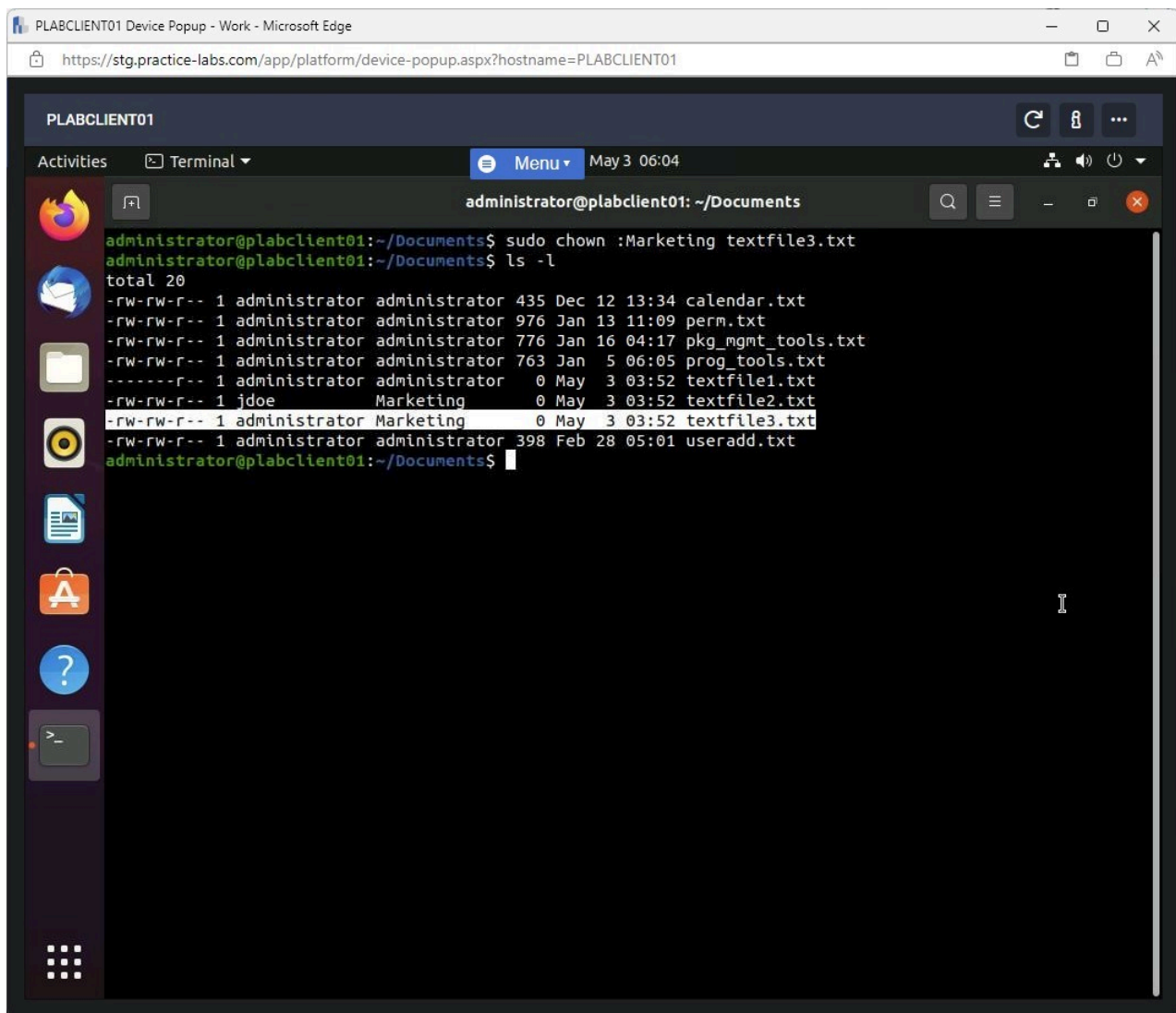


Figure 1.24 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **chown** command is now used to change the group of **textfile3.txt** to **Marketing**.

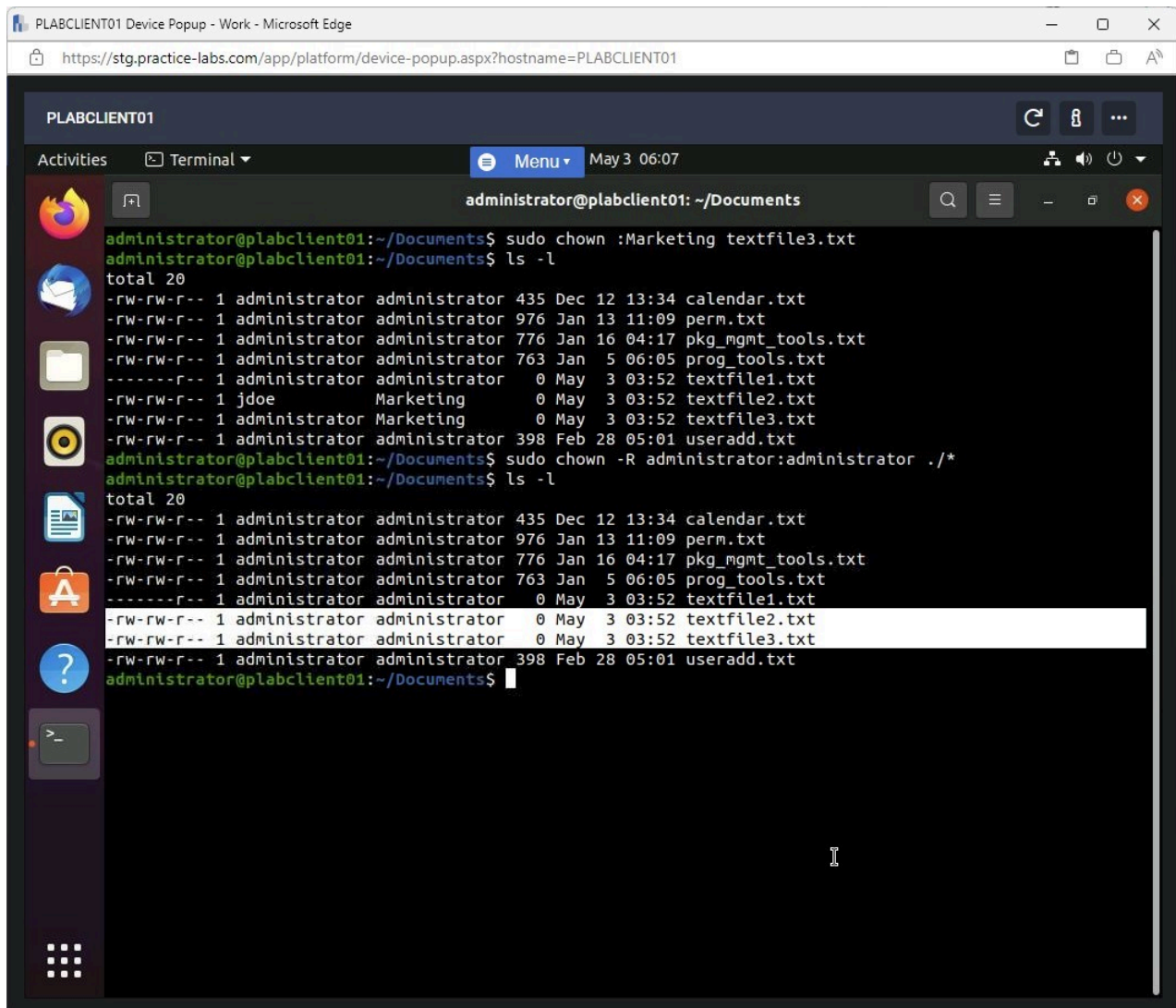
Step 15

In the **Terminal** window, type the following command and press **Enter**:

```
sudo chown -R administrator:administrator ./*
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a menu bar and a sidebar. The terminal output is as follows:

```
administrator@plabclient01: ~/Documents
administrator@plabclient01:~/Documents$ sudo chown :Marketing textfile3.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 jdoe Marketing 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator Marketing 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo chown -R administrator:administrator ./
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
-----r-- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 1.25 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above **chown** command will reset the ownership and the group of the files to the default settings. Notice that **jdoe** and **Marketing** are no longer assigned permissions.

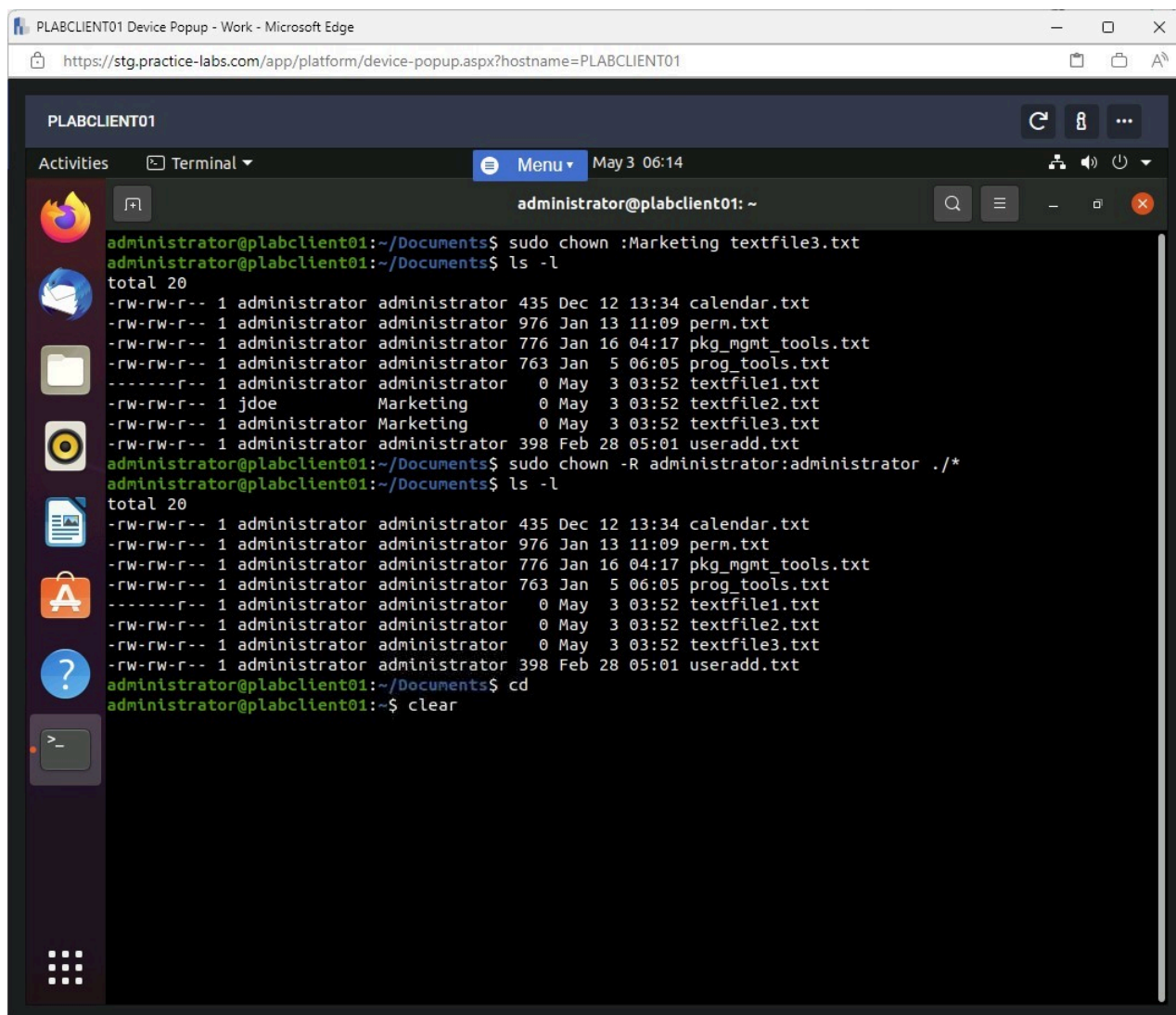
Step 16

In the **Terminal** window, type the following command and press **Enter**:

```
cd
```

Next, type the following command and press **Enter**:

clear



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal May 3 06:14
administrator@plabclient01: ~

administrator@plabclient01:~/Documents$ sudo chown :Marketing textfile3.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
----- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 jdoe Marketing 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator Marketing 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo chown -R administrator:administrator ./
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
----- 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ cd
administrator@plabclient01:~$ clear
```

Figure 1.26 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Keep the **Terminal** window open.

Exercise 2 - Controlling Access to Directory and Files with Advanced Permissions

Access Control Lists (ACLs) are used in Linux to give permissions to a user or group to access a file or a directory without changing the permission set of the file/directory. Different

permissions sets can be applied according to the type of access the user or group requires.

In this exercise, you will create users and groups. You will then configure permissions for users and groups with `getfacl` and `setfacl` and then configure the sticky bit.

Learning Outcomes

After completing this exercise, you should be able to:

- Configure Permissions using Setfacl and Create Inheritable Permission
- Set Sticky Bit to Control Access

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Configure Permissions using Setfacl and Create Inheritable Permissions

In this task, you will configure permissions using `setfacl` and create inheritable permissions.

Step 1

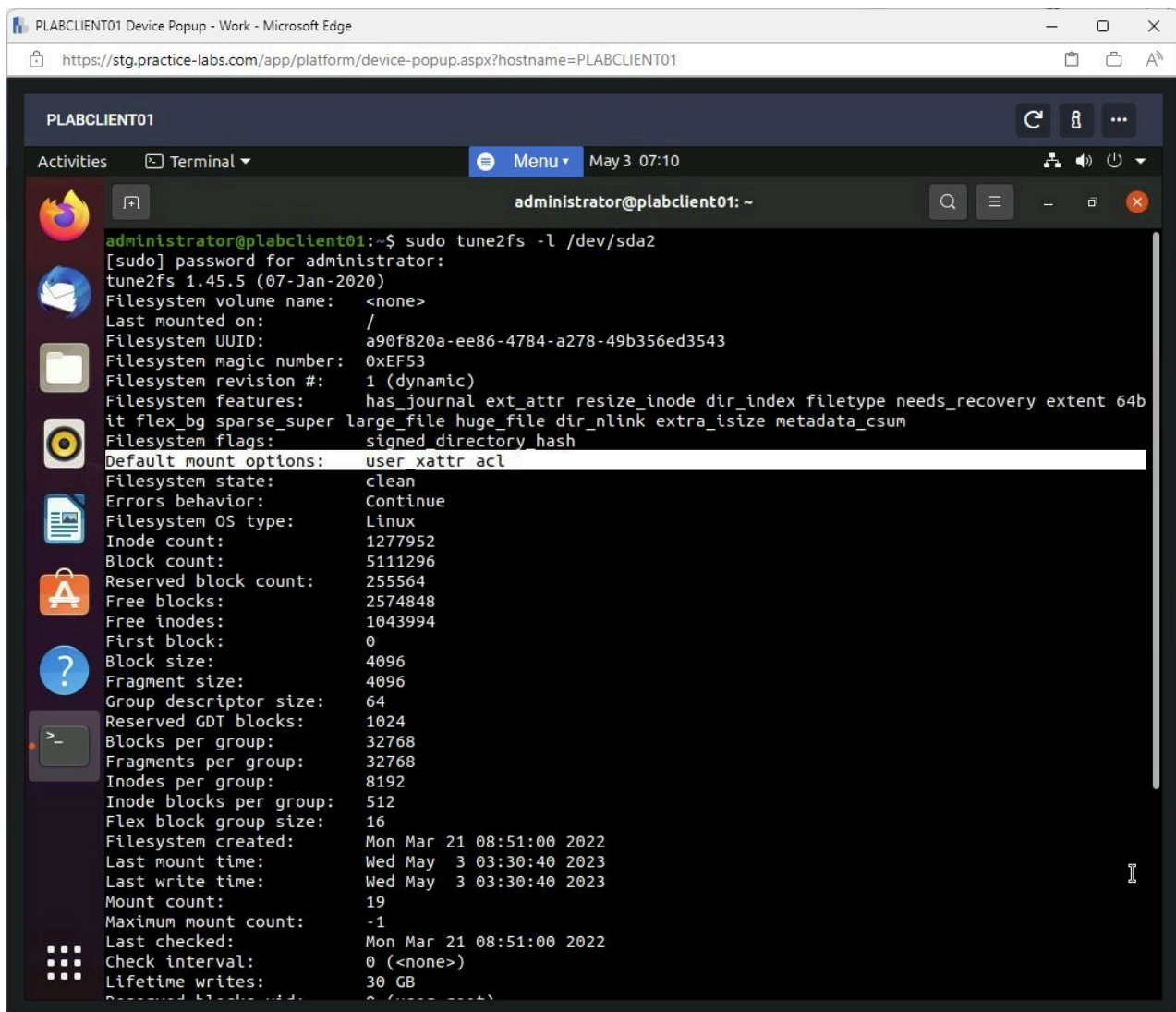
Ensure you are connected to **PLABCLIENT01** and the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo tune2fs -l /dev/sda2
```

If required, type the following **Password** and press **Enter**:

Passw0rd



The screenshot shows a terminal window titled 'PLABCLIENT01' with the prompt 'administrator@plabclient01: ~'. The command 'sudo tune2fs -l /dev/sda2' has been executed, displaying detailed filesystem information for /dev/sda2. The output includes the filesystem type (tune2fs 1.45.5), volume name, UUID, magic number, revision, features, flags, and various statistics. A key line in the output is 'Default mount options: user_xattr acl', which is highlighted in the image.

```
administrator@plabclient01:~$ sudo tune2fs -l /dev/sda2
[sudo] password for administrator:
tune2fs 1.45.5 (07-Jan-2020)
Filesystem volume name: <none>
Last mounted on: /
Filesystem UUID: a90f820a-ee86-4784-a278-49b356ed3543
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64b
it flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags: signed directory hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 1277952
Block count: 5111296
Reserved block count: 255564
Free blocks: 2574848
Free inodes: 1043994
First block: 0
Block size: 4096
Fragment size: 4096
Group descriptor size: 64
Reserved GDT blocks: 1024
Blocks per group: 32768
Fragments per group: 32768
Inodes per group: 8192
Inode blocks per group: 512
Flex block group size: 16
Filesystem created: Mon Mar 21 08:51:00 2022
Last mount time: Wed May 3 03:30:40 2023
Last write time: Wed May 3 03:30:40 2023
Mount count: 19
Maximum mount count: -1
Last checked: Mon Mar 21 08:51:00 2022
Check interval: 0 (<none>)
Lifetime writes: 30 GB
```

Figure 2.1 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above command is used to verify if **FACLs** (File Access Control Lists) can be used. Notice the line - Default mount options: user_xattr **acl**. The file is mounted with the **acl** option.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
cd Documents
```

Next, type the following command and press **Enter**:

```
sudo useradd leroybrown
```

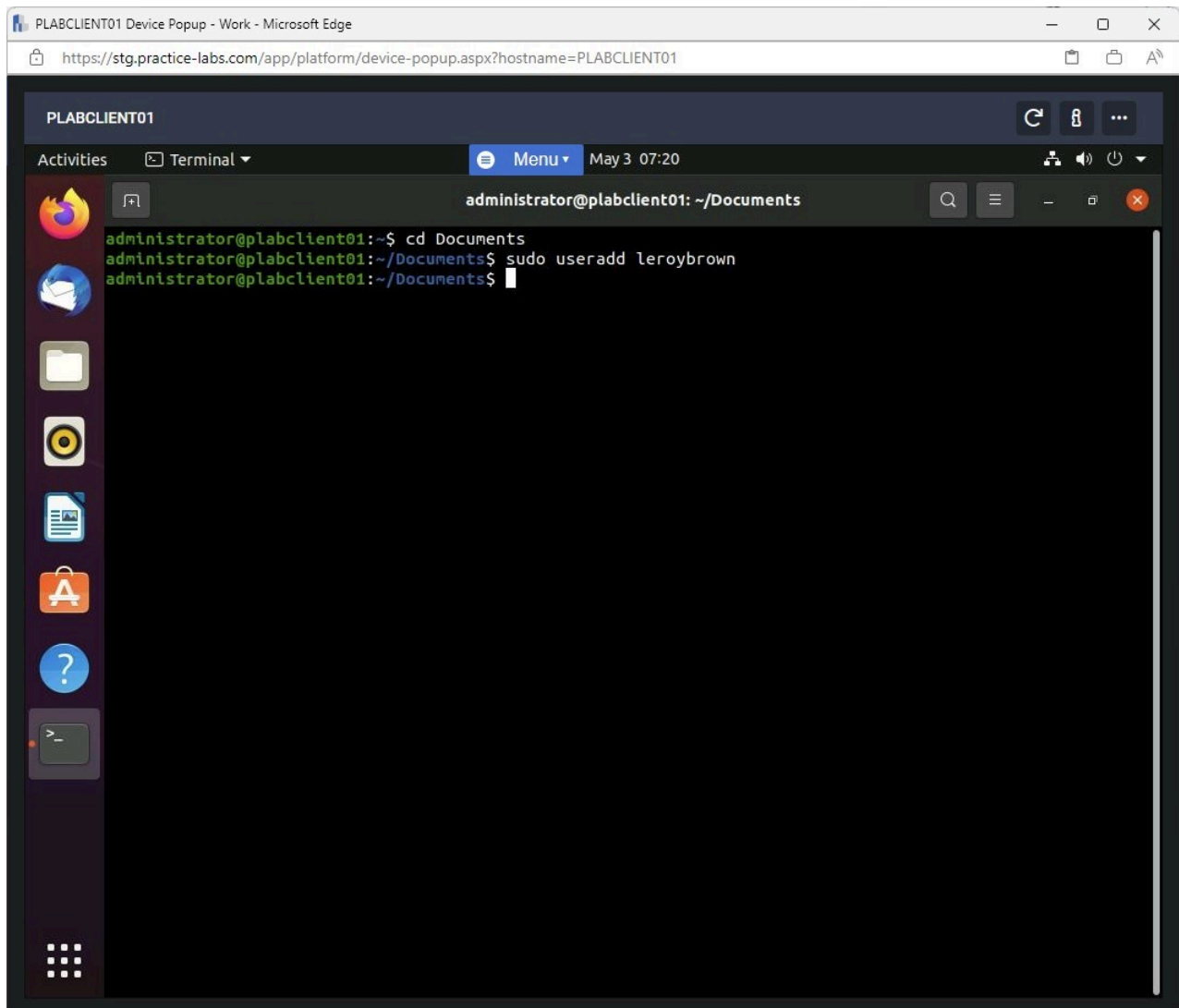


Figure 2.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

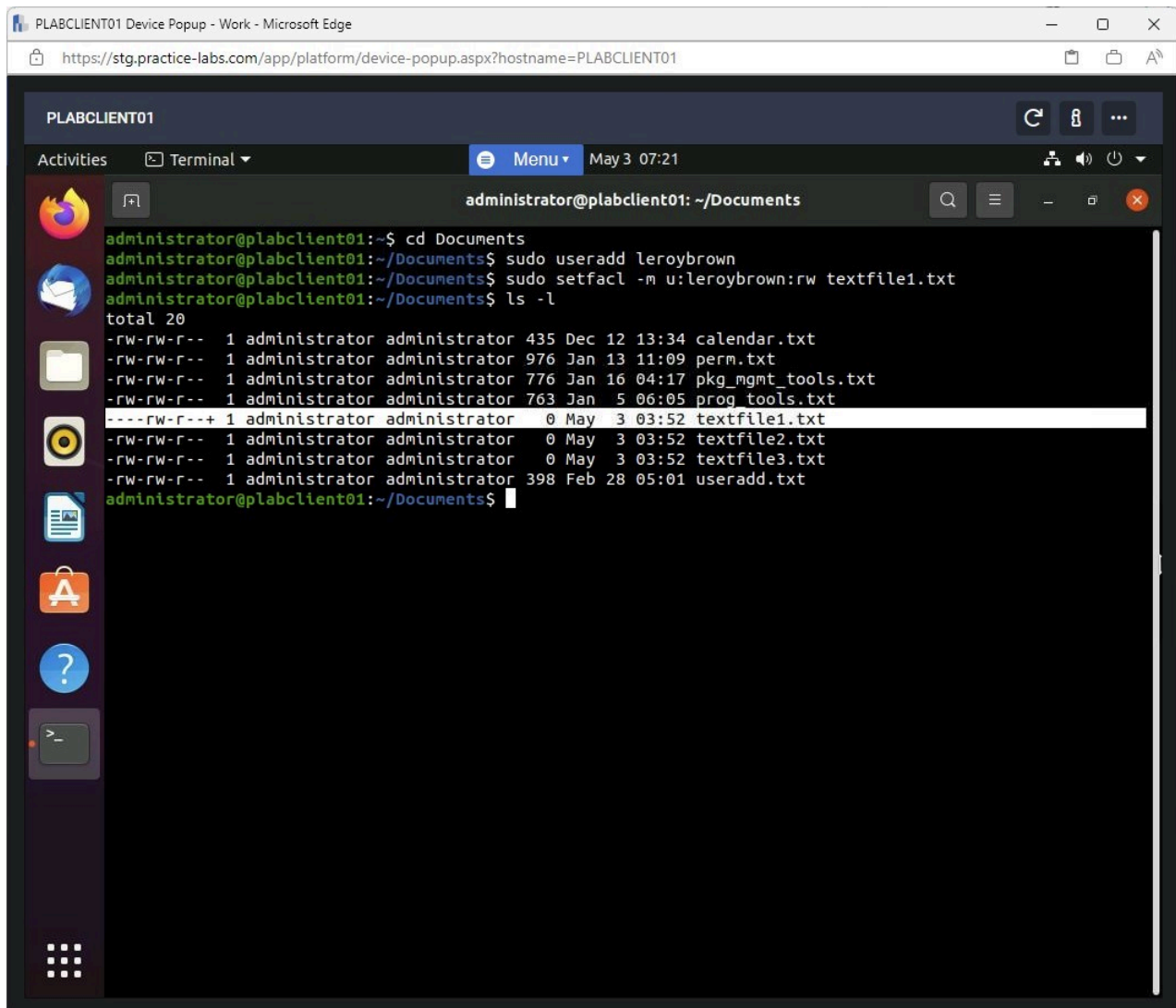
Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -m u:leroybrown:rw textfile1.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01" with the following commands and output:

```
administrator@plabclient01:~$ cd Documents
administrator@plabclient01:~/Documents$ sudo useradd leroybrown
administrator@plabclient01:~/Documents$ sudo setfacl -m u:leroybrown:rw textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
```

The output of the `ls -l` command is as follows:

```
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
---rw-r--+ 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
```

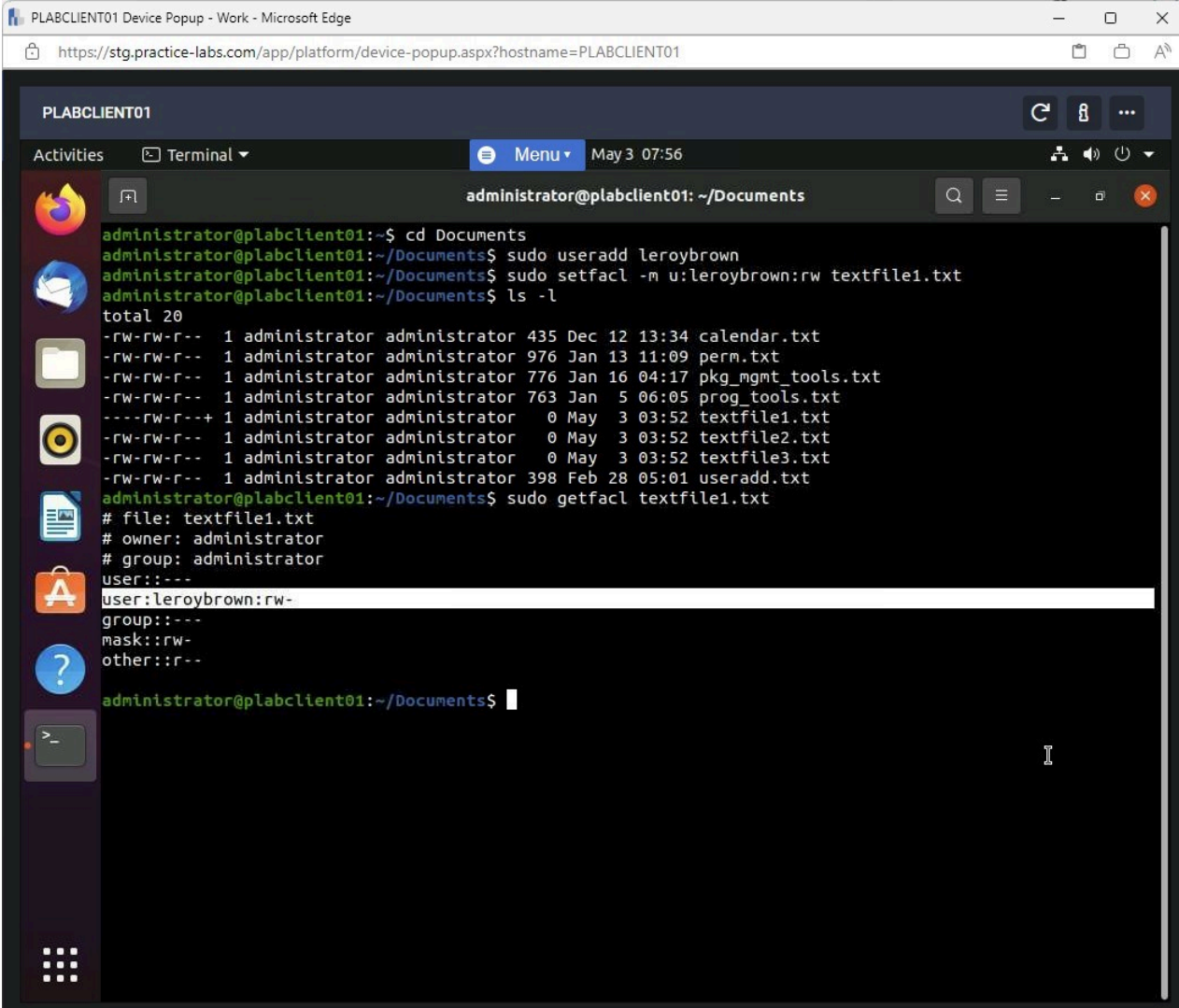
Figure 2.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The `setfacl` command is used to set, modify or remove ACLs for files and directories. The `-m` parameter will modify the current ACL of the file. Notice the `+` character. It indicates there is an ACL applied to the file. The user `leroybrown` is given read and write permissions.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo getfacl textfile1.txt
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with the following commands and output:

```
administrator@plabclient01: ~/Documents
administrator@plabclient01:~$ cd Documents
administrator@plabclient01:~/Documents$ sudo useradd leroybrown
administrator@plabclient01:~/Documents$ sudo setfacl -m u:leroybrown:rw textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
----rw-r--+ 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo getfacl textfile1.txt
# file: textfile1.txt
# owner: administrator
# group: administrator
user::---
user:leroybrown:rw-
group::---
mask::rw-
other::r--
administrator@plabclient01:~/Documents$
```

Figure 2.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **getfacl** command will display the file name, owner, group and the ACL. File, owner and group are known as the base ACL entries. Notice user **leroybrown** is the ACL associated with this file.

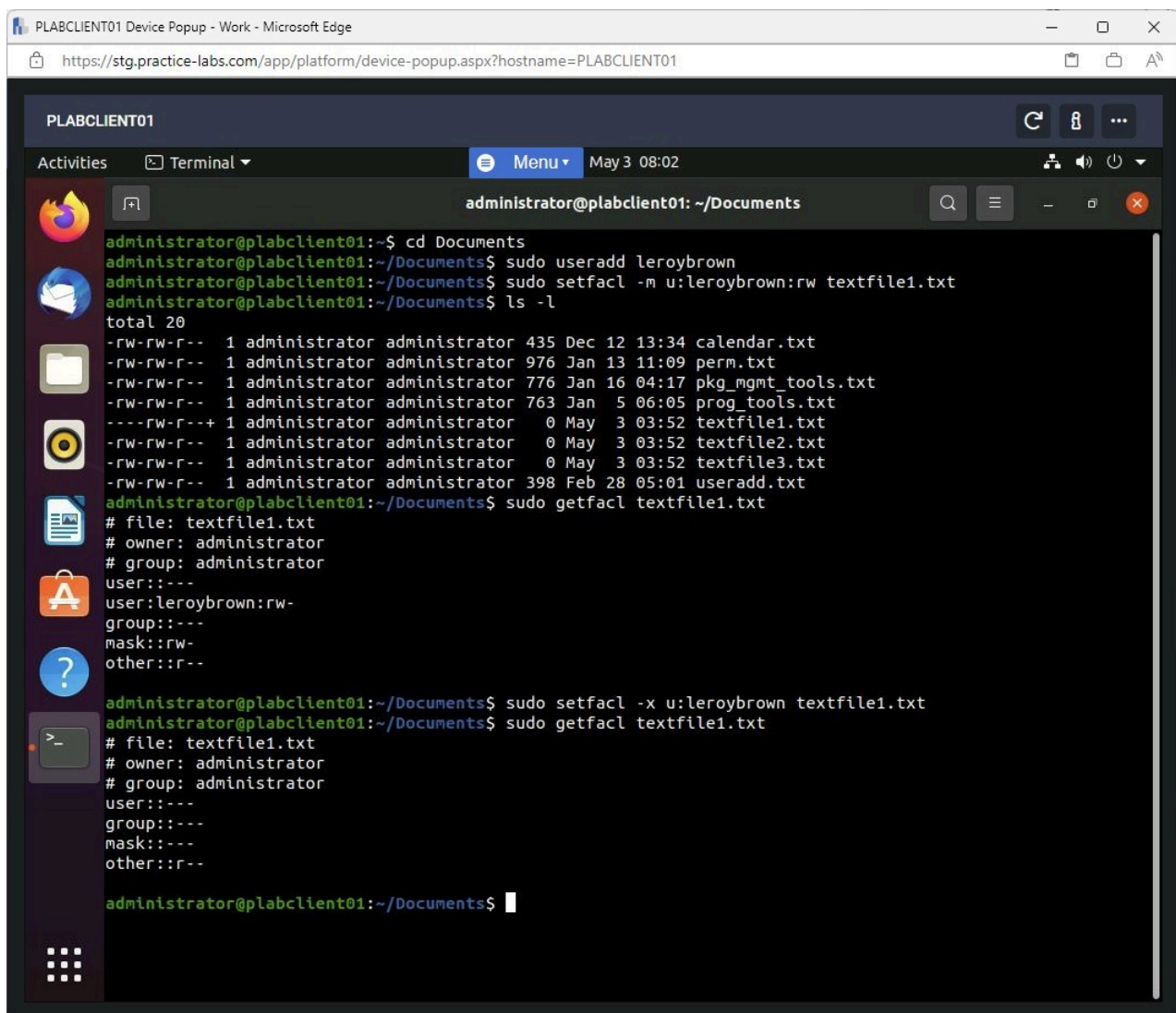
Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -x u:leroybrown textfile1.txt
```

Next, type the following command and press **Enter**:

```
sudo getfacl textfile1.txt
```



The screenshot shows a terminal window titled "PLABCLIENT01" with a dark background. The user is logged in as "administrator@plabclient01" in the directory "~/Documents". The terminal shows the following commands and output:

```
administrator@plabclient01:~/Documents$ cd Documents
administrator@plabclient01:~/Documents$ sudo useradd leroybrown
administrator@plabclient01:~/Documents$ sudo setfacl -m u:leroybrown:rw textfile1.txt
administrator@plabclient01:~/Documents$ ls -l
total 20
-rw-rw-r-- 1 administrator administrator 435 Dec 12 13:34 calendar.txt
-rw-rw-r-- 1 administrator administrator 976 Jan 13 11:09 perm.txt
-rw-rw-r-- 1 administrator administrator 776 Jan 16 04:17 pkg_mgmt_tools.txt
-rw-rw-r-- 1 administrator administrator 763 Jan 5 06:05 prog_tools.txt
----rw-r--+ 1 administrator administrator 0 May 3 03:52 textfile1.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile2.txt
-rw-rw-r-- 1 administrator administrator 0 May 3 03:52 textfile3.txt
-rw-rw-r-- 1 administrator administrator 398 Feb 28 05:01 useradd.txt
administrator@plabclient01:~/Documents$ sudo getfacl textfile1.txt
# file: textfile1.txt
# owner: administrator
# group: administrator
user::---
user:leroybrown:rw-
group::---
mask::rw-
other::r--

administrator@plabclient01:~/Documents$ sudo setfacl -x u:leroybrown textfile1.txt
administrator@plabclient01:~/Documents$ sudo getfacl textfile1.txt
# file: textfile1.txt
# owner: administrator
# group: administrator
user::---
group::---
mask::---
other::r--

administrator@plabclient01:~/Documents$
```

Figure 2.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **setfacl** command, when used with the **x** parameter, will remove the ACL entry for **leroybrown** from the current file. Notice the output of the **getfacl** command. The ACL entry for **leroybrown** is removed.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -m g:Marketing:rw textfile1.txt
```

Next, type the following command and press **Enter**:

```
sudo getfacl textfile1.txt
```

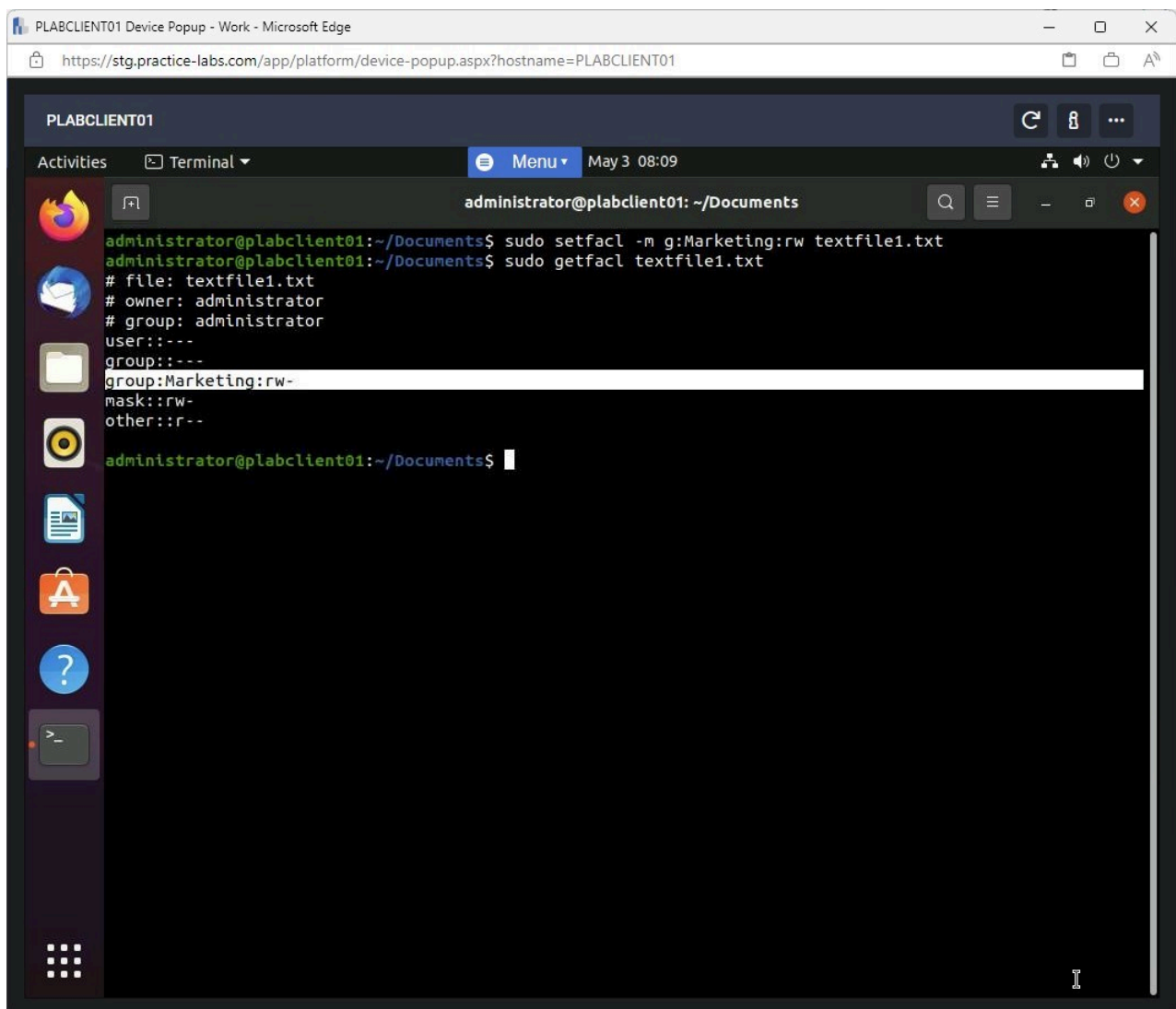


Figure 2.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **setfacl** command will configure permissions from the current ACL of the file for the **Marketing** group. Notice the entry for the **Marketing** group.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
cd
```

Next, type the following command and press **Enter**:

```
mkdir Reports
```

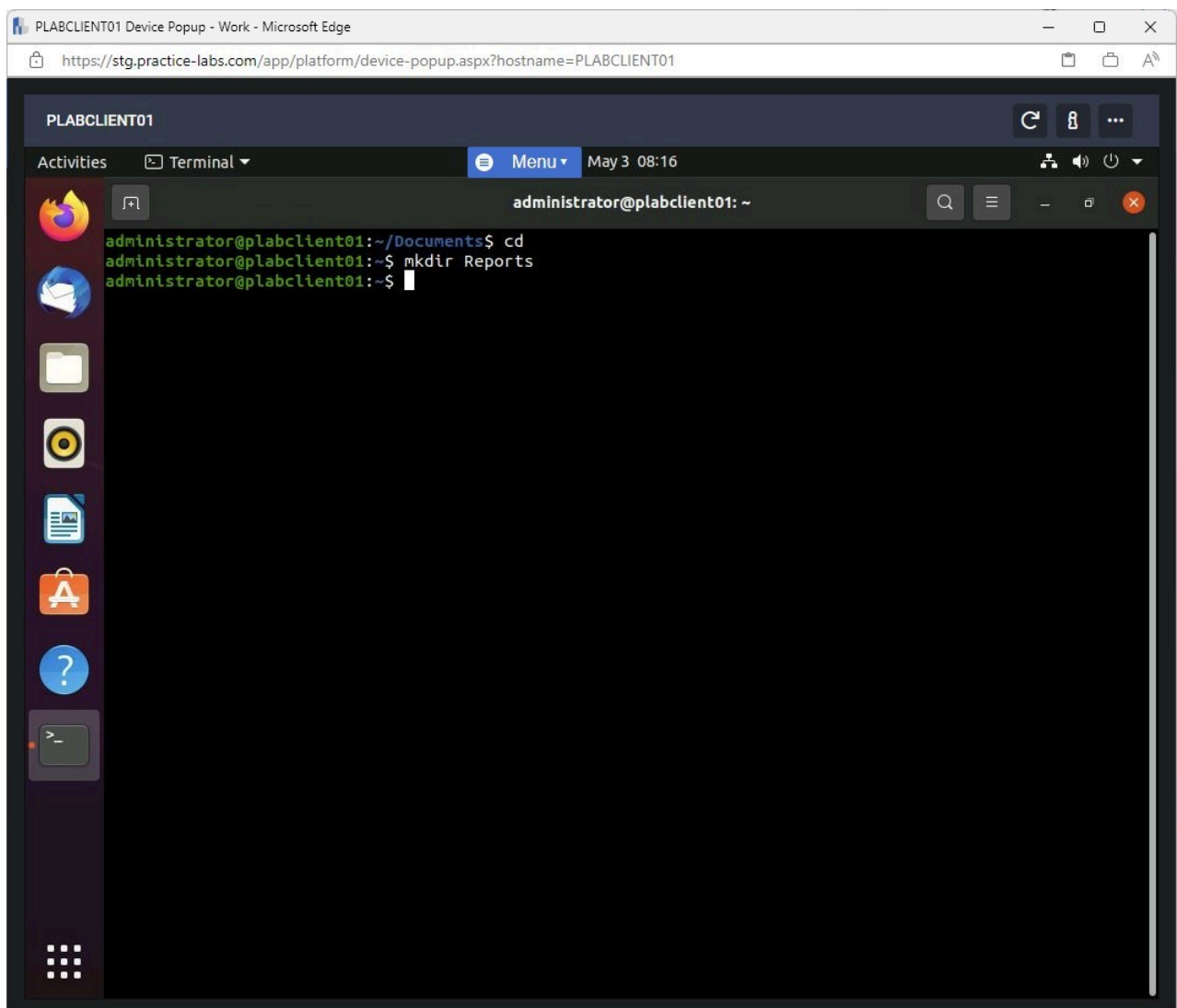


Figure 2.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *mkdir* command is used to create the **Reports** directory.

Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -m u:leroybrown:rw Reports
```

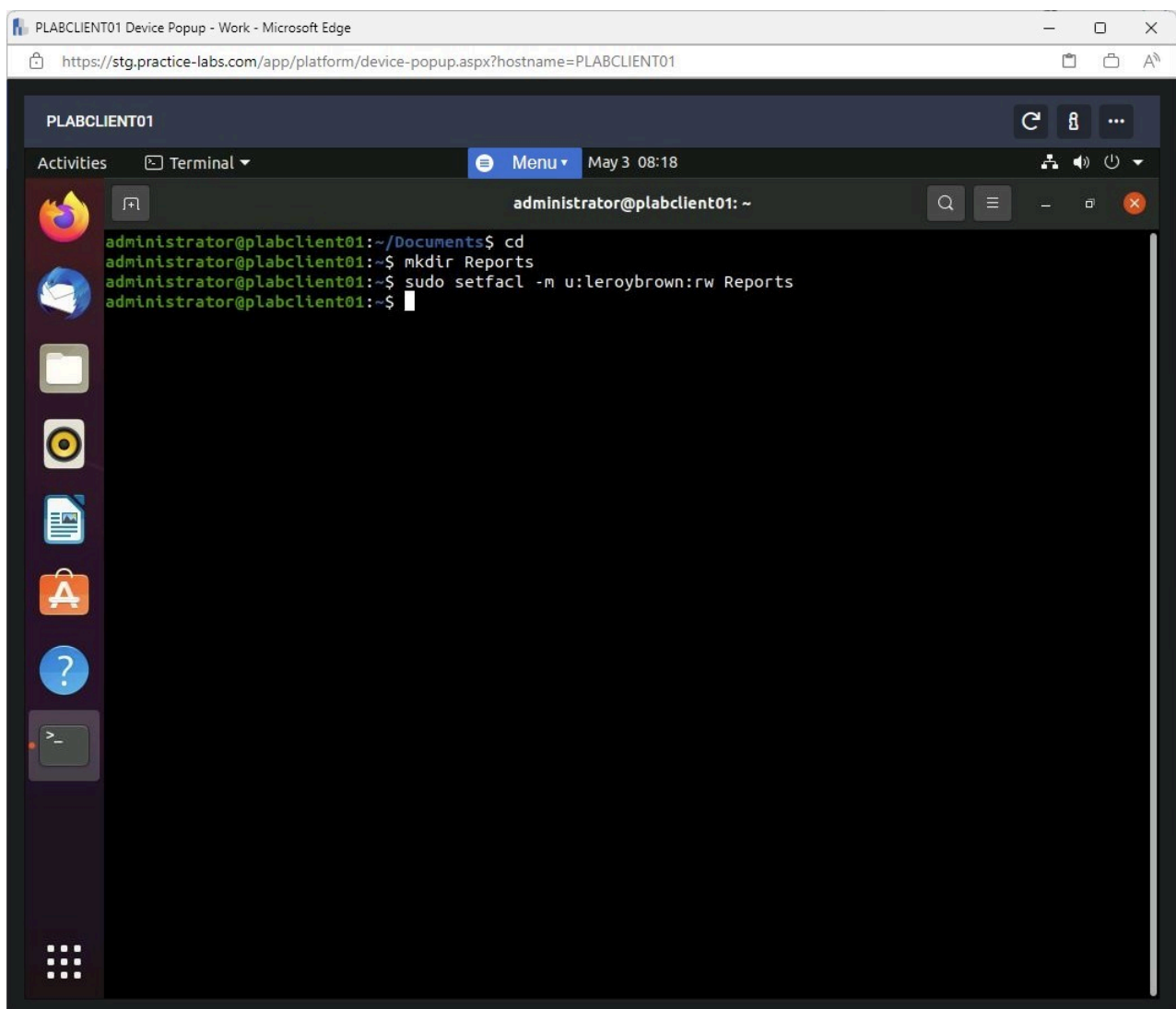


Figure 2.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The above command will modify the permissions for **leroybrown** and sets it in the **Reports** directory.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -m d:u:leroybrown:rw Reports
```

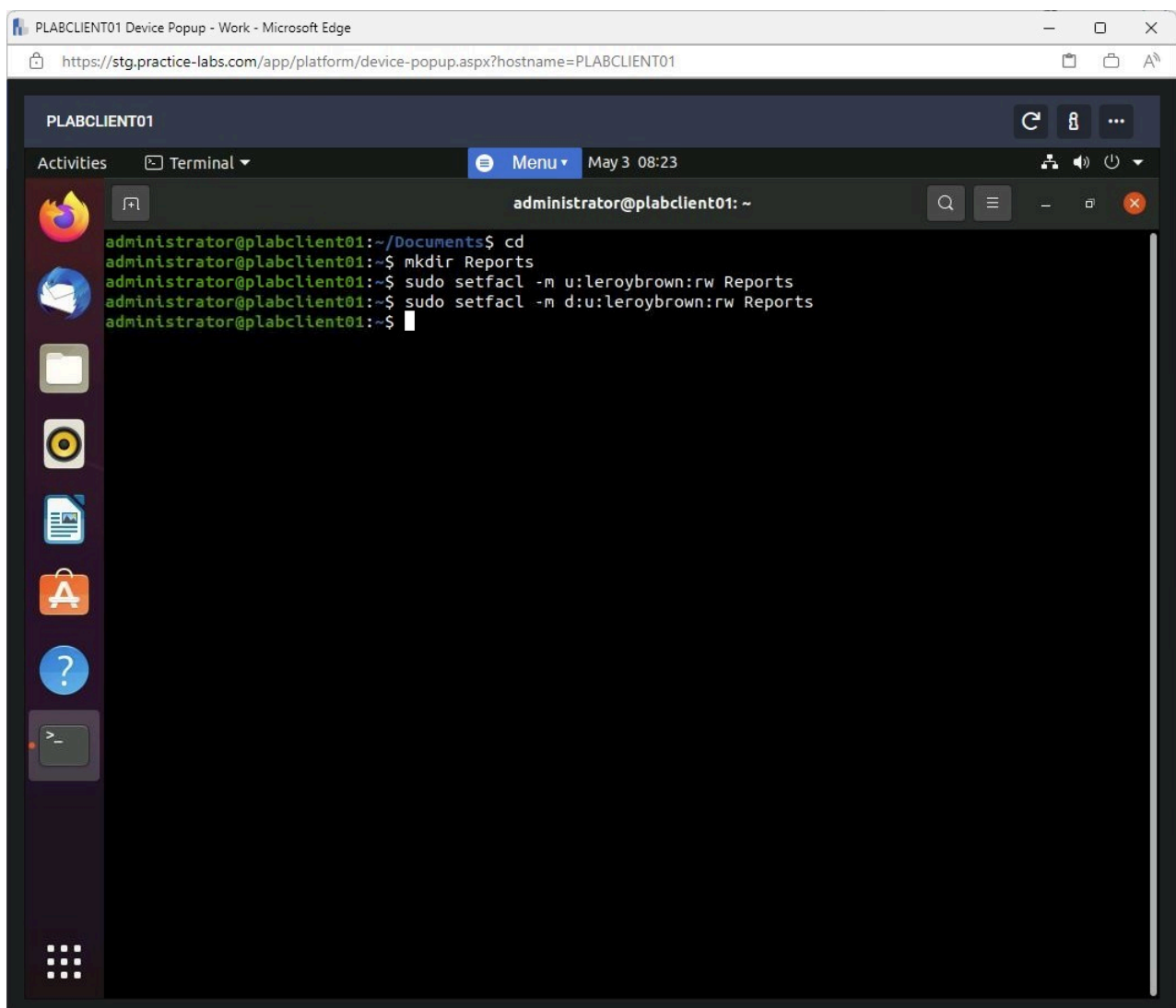


Figure 2.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: You have now set permission inheritance with the **setfacl** command for the user **leroybrown**. Permission Inheritance is when the permissions at the folder level get passed

onto objects within the file.

Step 10

In the **Terminal** window, type the following command and press **Enter**:

```
sudo setfacl -m g:Marketing:rw Reports
```

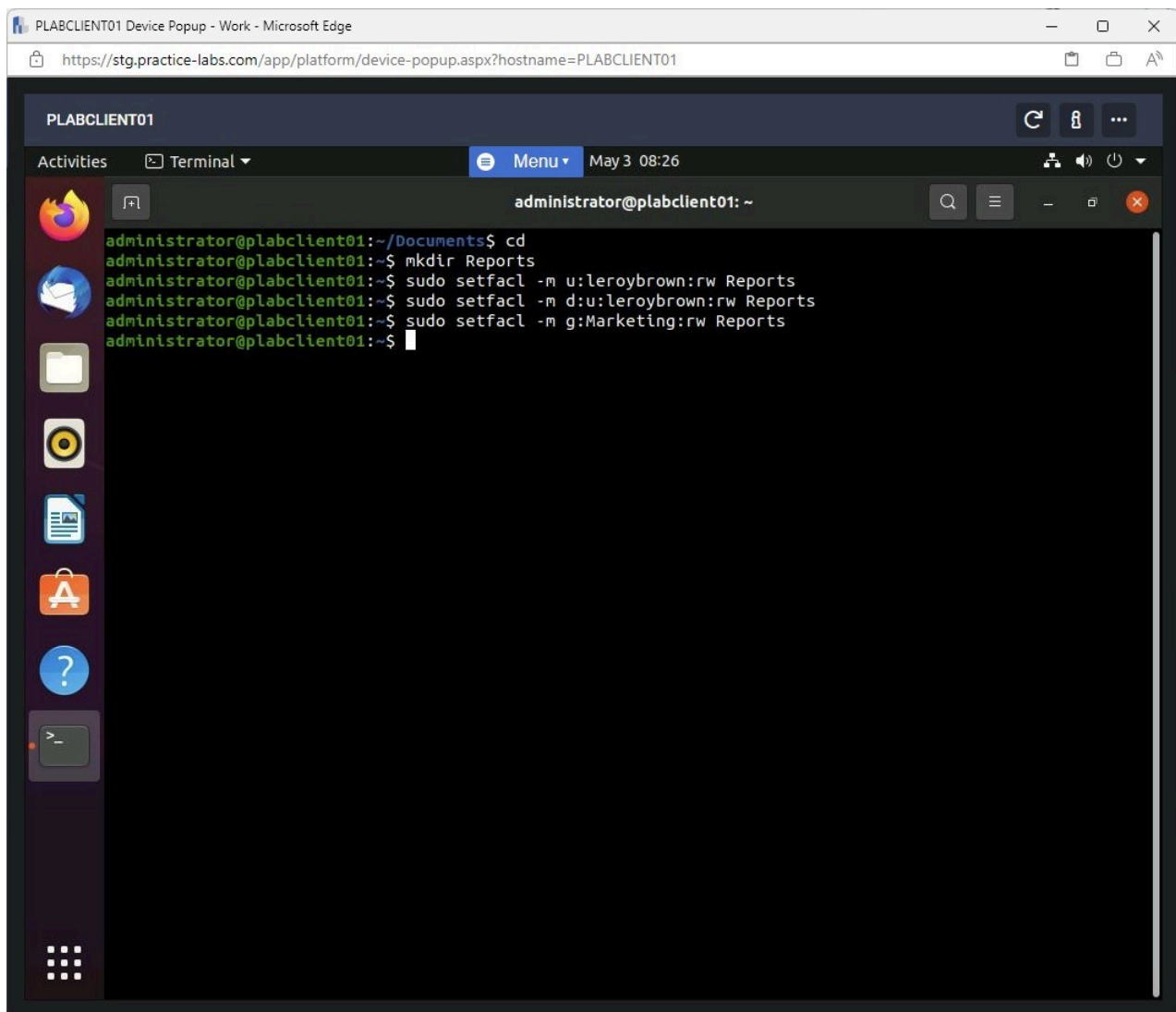


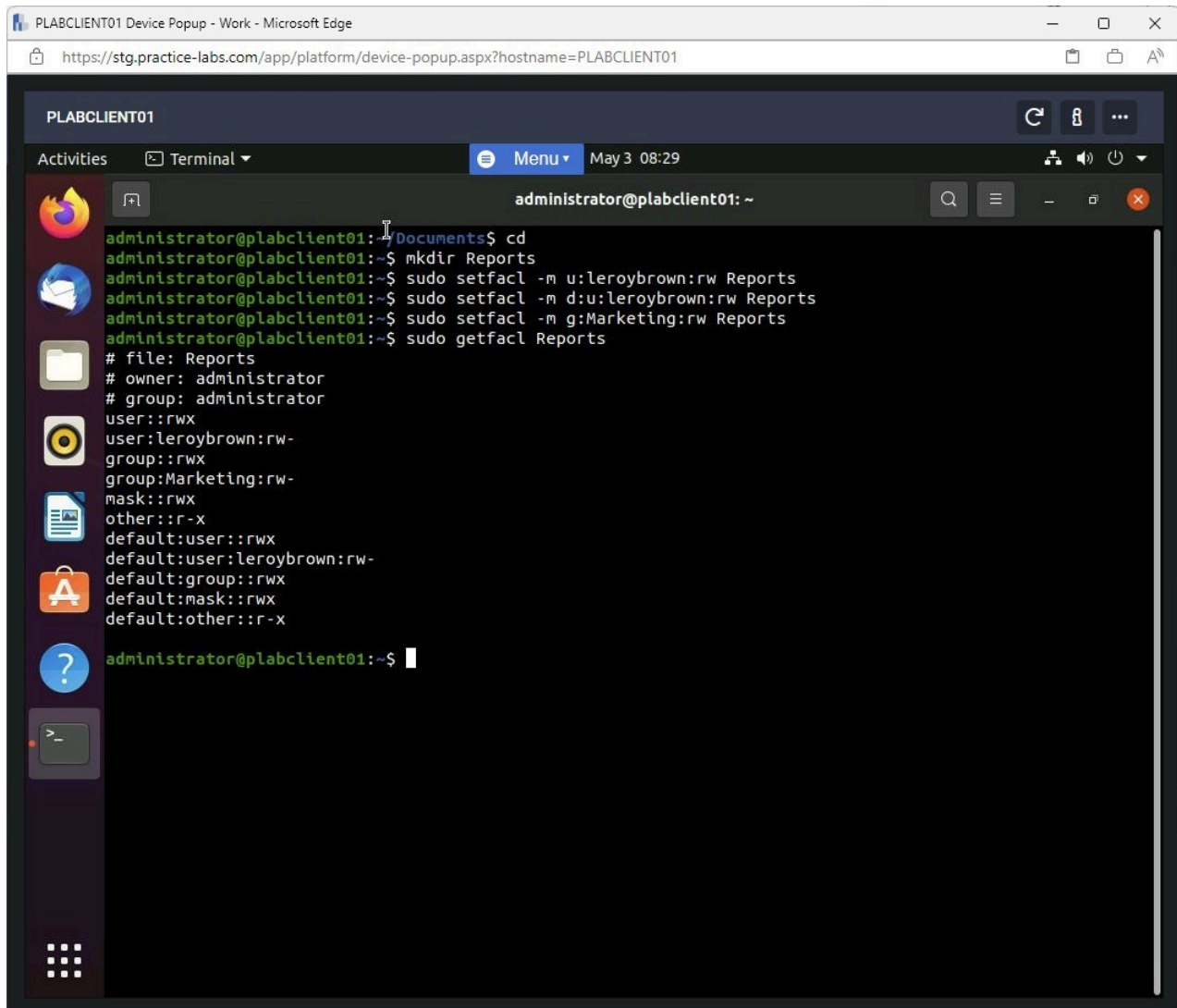
Figure 2.10 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *setfacl* command will set permission inheritance for the **Marketing** group.

Step 11

In the **Terminal** window, type the following command and press **Enter**:

```
sudo getfacl Reports
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 3 08:29
administrator@plabclient01: ~
administrator@plabclient01:~Documents$ cd
administrator@plabclient01:~$ mkdir Reports
administrator@plabclient01:~$ sudo setfacl -m u:leroybrown:rw Reports
administrator@plabclient01:~$ sudo setfacl -m d:u:leroybrown:rw Reports
administrator@plabclient01:~$ sudo setfacl -m g:Marketing:rw Reports
administrator@plabclient01:~$ sudo getfacl Reports
# file: Reports
# owner: administrator
# group: administrator
user::rwx
user:leroybrown:rw-
group::rwx
group:Marketing:rw-
mask::rwx
other::r-x
default:user::rwx
default:user:leroybrown:rw-
default:group::rwx
default:mask::rwx
default:other::r-x
administrator@plabclient01:~$
```

Figure 2.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice from the output both the user **leroybrown** and the group **Marketing** have been set permission inheritance.

Keep the **Terminal** window open.

Task 2 - Set Sticky Bit to Control Access

A sticky bit permission is set on directories to ensure that only the root users can modify the contents of the directory.

In this task, you will create a directory and then modify permissions to limit users from manipulating files they do not own.

Step 1

Connect to **PLABCLIENT01**, where the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
mkdir Data
```

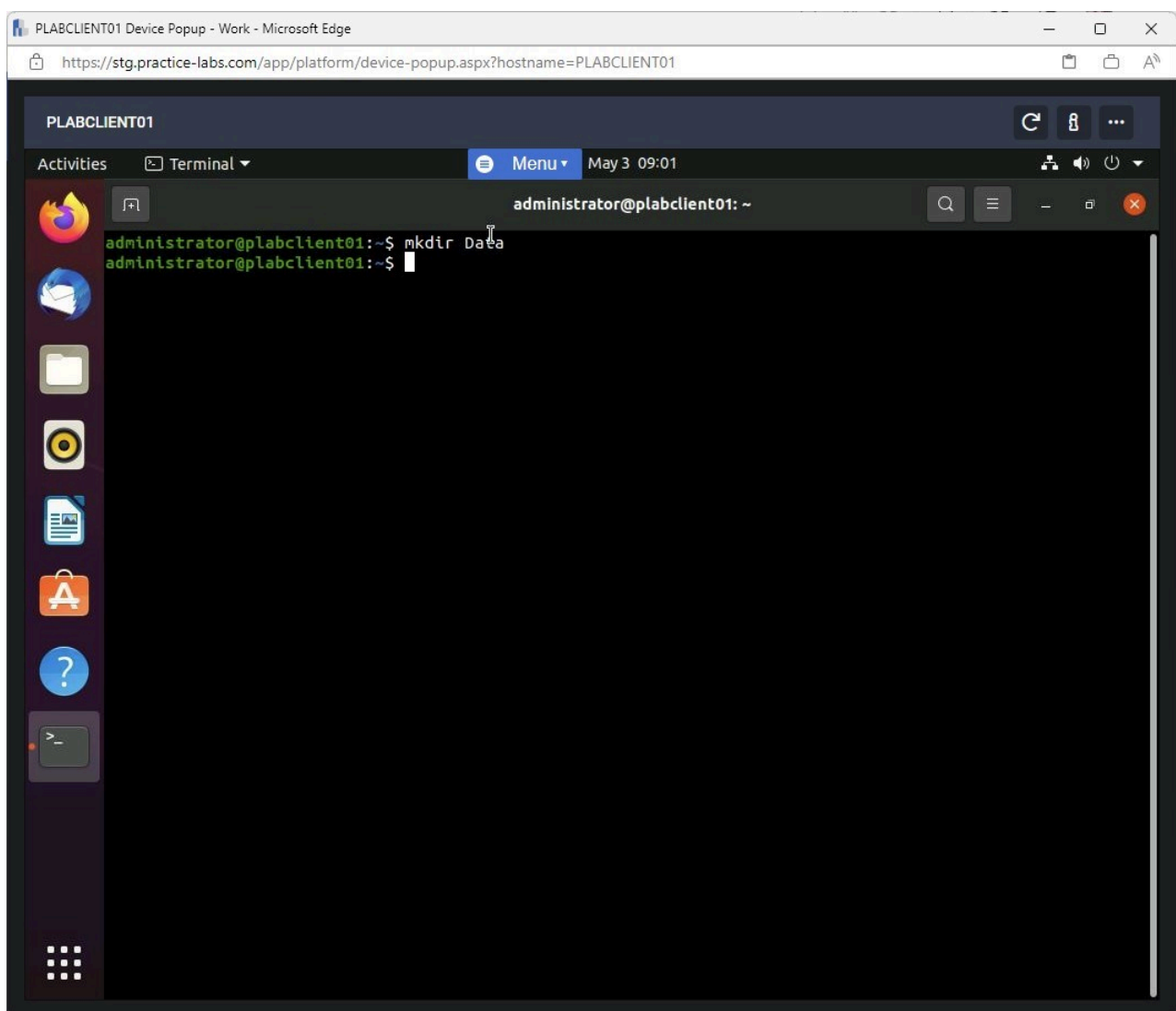


Figure 2.12 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
sudo chmod o+t Data
```

***Note:** If required, type the following Password and press Enter:*

Passw0rd

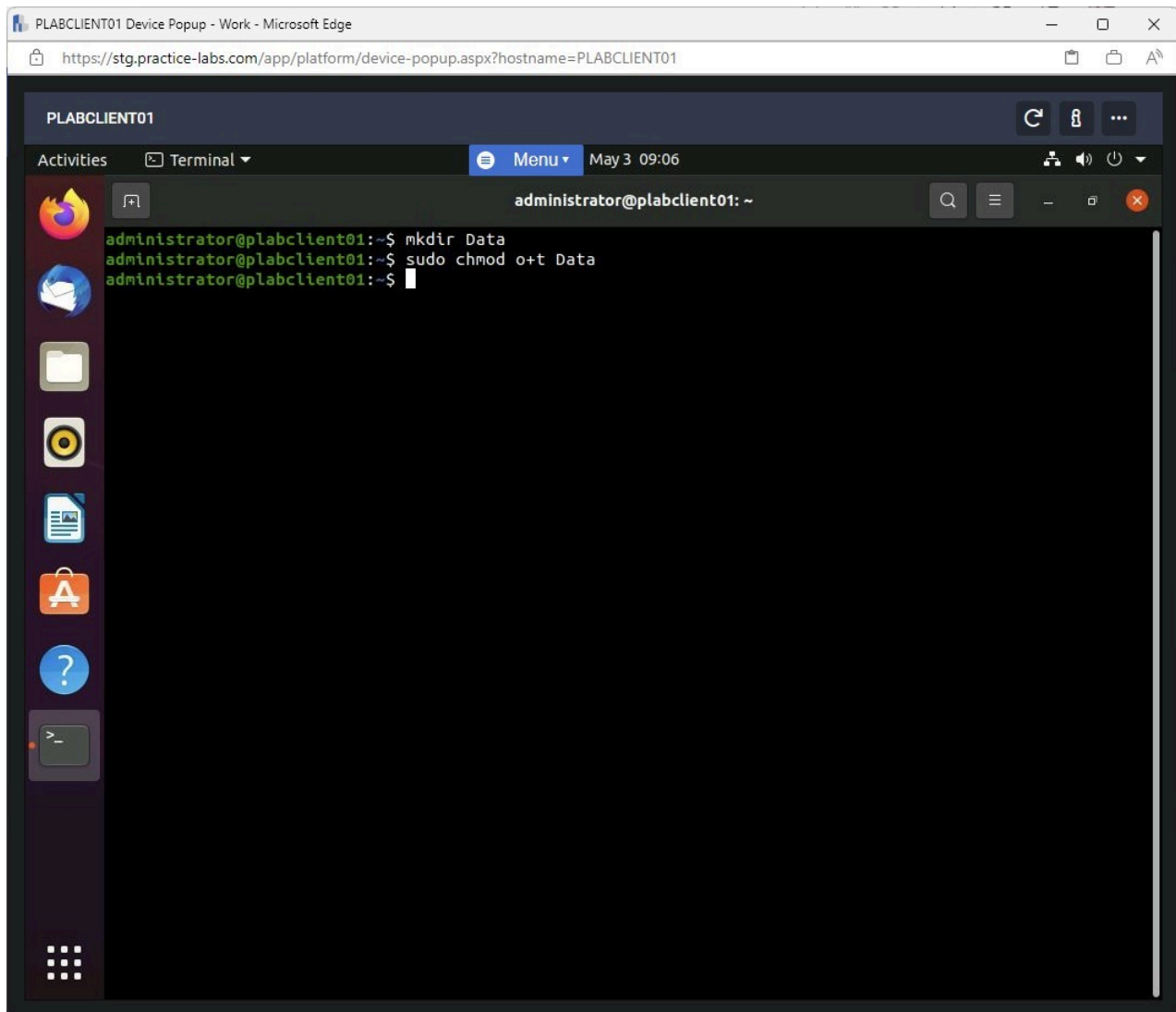


Figure 2.13 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

***Note:** The **chmod** command is used with the **t** flag to set a sticky bit permission for other (o) users.*

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

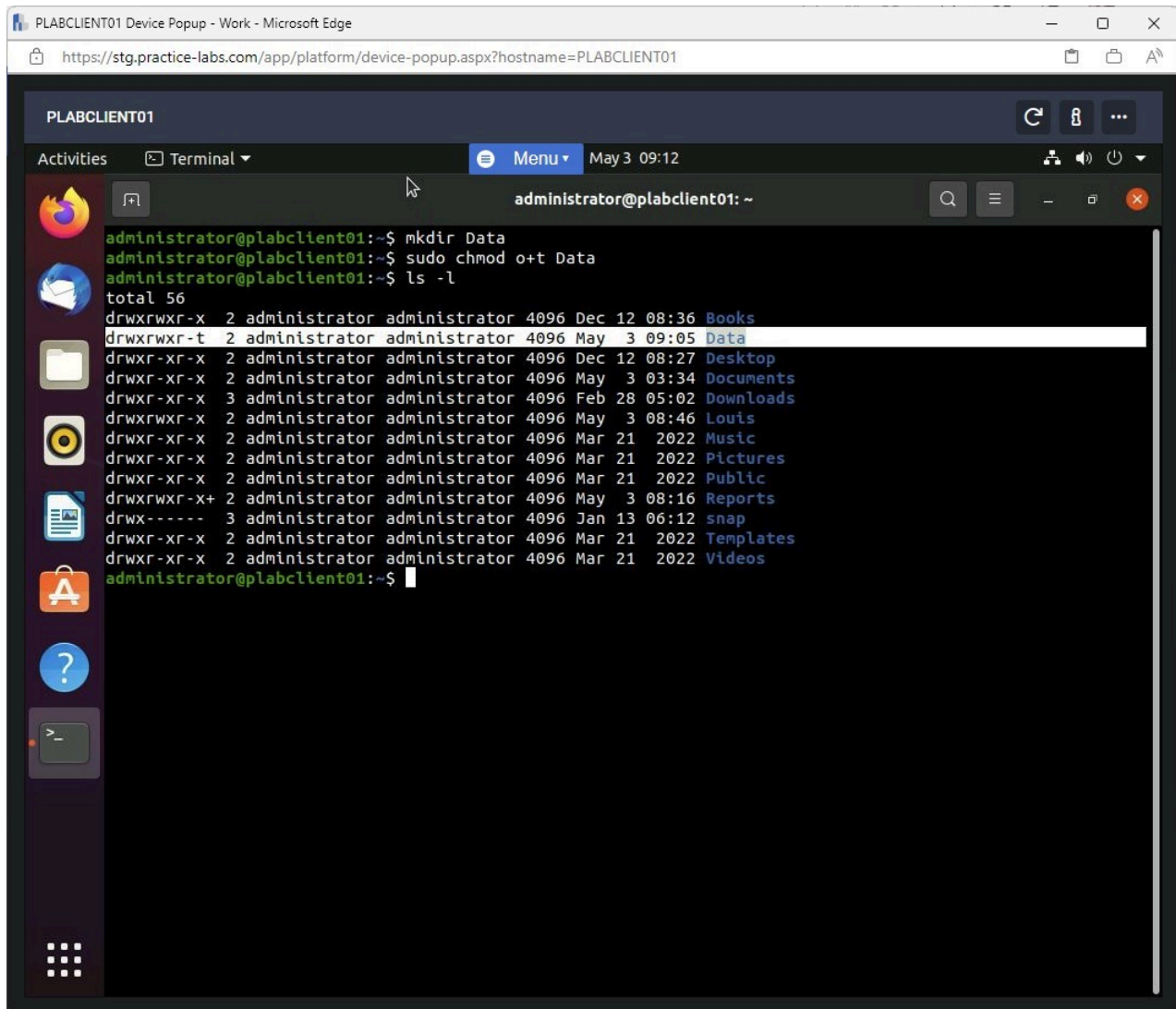


Figure 2.14 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the *t* at the end representing a sticky bit permission. Sticky bit permissions are ideal when using shared storage.

Close the **Terminal** window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Access Control Utilities** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Controlling Access to Directory and Files with Standard Permissions
- Exercise 2 - Controlling Access to Directory and Files with Advanced Permissions

You should now be able to:

- Create Files and View Permissions
- Modify File Permissions
- Configure Permissions using Setfacl and Create Inheritable Permission
- Set Sticky Bit to Control Access

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.